



Aspect-Based Sentiment Analysis of Code-Switched Consumer Reviews for Local Coffee and Matcha Businesses in Davao City

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Chapter 1

Introduction

1.1 Background of the Study

Cafe culture, particularly matcha and coffee shop hopping, became increasingly popular in the Philippines. This trend led to a significant rise in demand for Matcha and Coffee, prompting multiple shops to open in the city. In urban areas such as Davao City, this rapid increase has led to a highly competitive and saturated market [20]. Consequently, consumers are given a hard time efficiently comparing cafés. This concern becomes even more significant as people from different parts of the country travel to Davao City specifically to explore its growing coffee and matcha scene.

Despite the availability of online reviews on platforms such as Google Maps, Facebook, and Instagram, these reviews are often unstructured, not normalized, fragmented, and overwhelming in volume. The presence of many positive reviews and high rankings for a business or its products may make it more likely that a potential consumer will feel confident about making a purchase [28]. However, the large volume of reviews requires consumers to manually sift through them, making decision-making time-consuming and inefficient. Furthermore, some of these reviews may contain biased, misleading, or even irrelevant information, which can negatively affect consumer choices and business reputations [30].

This major limitation stems from the inability of existing systems to effectively organize, analyze, and interpret large volumes of unstructured review data. Although thousands of reviews are available across platforms, consumers must manually read them in raw, unstructured text, which is both time-consuming and inefficient. Moreover, this issue is further complicated by code-switching, informal language, and inconsistent expressions, making it difficult, and in some cases impossible, for traditional analysis methods to accurately capture sentiment and meaning [36][31].

To address these challenges, prior studies have used automated data collection methods, such as Application Programming Interfaces (APIs), to convert raw reviews into structured datasets for analysis, thereby improving scalability and data reliability [39]. Sentiment analysis, on the other hand, relied on lexicon-based techniques such as

VADER, most especially in the earlier years [8]. Although it was useful in English-written texts, the techniques were found to be limited in handling multilingual and code-switched texts commonly found in Philippine contexts. As a result, more advanced learning-based models, particularly multilingual transformer models such as XLM-RoBERTa, were introduced to better interpret complex language patterns and improve classification performance [36]. Multiple studies have also utilized aspect-level analysis through Aspect-based Sentiment Analysis (ABSA) to extract more detailed insights by evaluating specific attributes [53]. Extensive manual annotation was also greatly reduced by using hybrid strategies, including Active learning, which has been found to improve performance [56]. The aforementioned approaches offer several advantages for addressing the limitations of traditional text and review analysis, particularly when integrated into a unified process. While standard Application Programming Interfaces (APIs) offer structured data, they often impose severe extraction limits. Thus, customizing an automated web scraping process ensures greater scalability and sufficient data volume when handling large datasets.

Furthermore, the application of Aspect-Based Sentiment Analysis (ABSA) enables the decomposition of reviews into distinct attributes, yielding more precise insights. To improve efficiency and reduce reliance on large labeled datasets, techniques such as Active Learning prioritize the most informative samples for annotation, significantly reducing labeling effort while maintaining performance [42]. Lastly, data augmentation strategies, such as regulated synthetic data generation, help address issues of data scarcity and class imbalance [22]. By using these approaches, a framework is created that transforms unstructured, multilingual, and context-rich texts into more meaningful insights.

Though the Davao City cafe and matcha shop scene is rapidly expanding, there is a distinct lack of research utilizing advanced Natural Language Processing (NLP) techniques to evaluate consumer preferences. The study fills the gap by implementing XLM-RoBERTa a Aspect-Based Sentiment Analysis (ABSA) framework, enhanced by Active Learning, specifically for Taglish and Bislish reviews.

1.2 Problem Statement

This study investigates the challenge of effectively analyzing unstructured, multilingual, and context-rich consumer reviews, specifically those that contain code-switching and informal language. Existing platforms generate large volumes of raw, unstructured textual data that is challenging to interpret using traditional methods, hindering the extraction of meaningful, reliable results. Specifically, this study aims to answer the following questions:

1. What aspect categories characterize the multilingual consumer reviews of local coffee and matcha businesses in Davao City, and how does the code-switched nature of the text affect aspect-level sentiment classification?
2. How does the proposed Multilingual Transformation model perform compared to lexicon-based approaches on code-switched and informal review texts?
3. What are the aspect categories that emerged from the consumer reviews of Davao City and matcha establishments, and what are the prevailing consumer sentiments associated with these discovered dimensions?

4. How can active learning and synthetic data generation improve model performance and reduce annotation effort?
5. How can the proposed model be evaluated and integrated into a system that supports consumer decision-making and business analysis?

1.3 Objectives

The main objective of this study is to develop an XLM-RoBERTa-based system for aspect-based sentiment analysis that efficiently handles multilingual and code-switched consumer reviews. The study also wishes to accomplish the following objectives:

1. To identify the aspect categories present in multilingual consumer reviews of local coffee and matcha businesses in Davao City and examine how code-switching affects aspect-level sentiment classification.
2. To compare the performance of the proposed XLM-RoBERTa-based model against lexicon-based approaches on code-switched and informal review texts.
3. To identify the aspect categories that emerged from consumer reviews of Davao City coffee and matcha establishments and determine the prevailing consumer sentiments associated with each discovered dimension.
4. To implement active learning strategies and synthetic data generation to reduce annotation effort while maintaining classification accuracy.
5. To evaluate the proposed model and integrate it into a system that supports consumer decision-making and business analysis.

1.4 Significance

This study presents significant contributions to various stakeholders by addressing the challenge of analyzing large volumes of unstructured, multilingual, and context-rich consumer reviews.

1. **For Consumers:** The rapid growth of cafe culture in Davao City has led to a highly competitive and saturated market [20]. Consequently, consumers face difficulties in efficiently comparing different establishments. An Aspect-Based Sentiment Analysis (ABSA) system alleviates the burden of manually processing thousands of reviews. Rather than reading extensive texts, consumers are provided with concise, data-driven summaries of specific aspects they find most valuable.
2. **For Local Coffee and Matcha Businesses:** Aggregated star ratings often fail to provide actionable insights. For instance, a single review may contain positive sentiments regarding a product's taste but negative sentiments toward the establishment's customer service or ambiance. Decomposing reviews into specific features enables businesses to accurately pinpoint operational strengths and identify specific areas requiring improvement.

3. **For Academic Research:** This study fine-tunes XLM-RoBERTa on Taglish and Bislish, code-switched dialects prevalent in the Philippines. Data scarcity constrains model performance; this work employs two complementary techniques to mitigate it. Active Learning identifies the most informative unlabeled examples for manual annotation based on model uncertainty [56]. Synthetic data generation then augments the training set, reducing the annotation burden while maintaining dataset quality.

1.5 Scope and Limitations

This study analyzes consumer reviews of local coffee and matcha businesses in Davao City. To ensure a comprehensive analysis, data collection encompasses all publicly available reviews for the selected establishments from their inception up to the date of extraction, with no minimum threshold or maximum limit imposed on the dataset volume. Due to the extraction limits of Google Places API, reviews are acquired through custom HTML scraping to ensure sufficient data volume and will be parsed into multiple CSV files, explicitly separated by coffee and matcha establishments. The dataset consists of multilingual, code-switched text spanning English, Tagalog, and Bisaya. A fine-tuned XLM-RoBERTa model performs aspect-based sentiment classification across predefined dimensions. Active learning is applied to identify uncertain samples for manual annotation, reducing labeling effort, while LLM-based synthetic data generation addresses class imbalance in underrepresented aspect-sentiment pairs.

Model collapse and potential loss of linguistic diversity resulting from synthetic data generation are not addressed in this study. Synthetic data are used only to supplement underrepresented classes, and human-annotated reviews remain the primary training source. Data collection is constrained by the inherent limitations of web scraping, such as dynamic website structure changes (DOM updates) and potential IP blocking. The finalized model will be integrated into a system. However, the underlying machine learning pipeline focuses exclusively on textual data. Image and video content in reviews are not processed.

1.6 Definition of Terms

1. **Active Learning (AL):** A machine learning framework that reduces manual annotation effort by selecting only the most informative or uncertain samples for labeling, rather than processing an entire dataset.
2. **Application Programming Interface (API):** A set of protocols that automates the extraction of large volumes of online consumer reviews into structured formats, such as CSV files, for further analysis.
3. **Aspect-Based Sentiment Analysis (ABSA):** A natural language processing technique that identifies specific attributes of a subject in text, such as taste, price, service, and ambiance, and determines the sentiment expressed toward each.

4. **Code-Switching:** The practice of alternating between two or more languages within a single text, as seen in “Taglish” (Tagalog-English) and “Bislish” (Bisaya-English), the dominant mixed-language variants in the dataset.
5. **Lexicon-Based Approaches:** Sentiment analysis methods that classify text using predefined word lists where each entry carries a positive or negative polarity score, with VADER as a representative example.
6. **Model Collapse:** A sustained decline in model performance caused by training predominantly on synthetic rather than human-generated text, resulting in reduced linguistic diversity over successive iterations.
7. **Multilingual BERT (XLM-RoBERTa):** A pre-trained transformer model that processes text across multiple languages in context, capable of interpreting code-mixed reviews written in Filipino dialects and English.
8. **Synthetic Data Generation:** The process of producing artificial training examples from a small seed set of real reviews using large language models, applied to address data scarcity and class imbalance.
9. **Unstructured Data:** Raw text lacking a predefined schema or organization, such as user-generated online reviews, which resists systematic processing without specialized tools.
10. **Web Scraping:** An automated technique that extracts data from websites by parsing HTML to gather raw online reviews at scale.

Chapter 2

Review of Related Literature

2.1 Local Coffee and Matcha Market

The rapid increase of cafe culture in the Philippines has significantly transformed local beverage shops [20]. Urban cities like Davao City have experienced exponential growth in coffee and matcha, creating a highly saturated, competitive environment. Local consumers, particularly the Gen Z demographic, are a key driver of this market.

1. *Market Competition:* The local coffee shop industry in Davao City has risen, evolving into a competitive service environment. When new establishments open, the market is quick to absorb them. However, sustaining the business requires distinct differentiations. Local cafe owners are leveraging locally sourced products (Mt. Apo coffee and award-winning Davao cacao) to establish their own unique market. This stimulates demand for both local and national brands and local competitors [3].
2. *Influencers:* Consumer decisions in Davao City are dependent on digital social proof [29]. Local consumers usually conduct online research before making purchasing decisions. Hence, consumer choices are heavily influenced by online reviews, social media recommendations, and peer opinions, making reputation management and digital influence necessary for market success in the region.
3. *Digital Engagement:* Gabato et al. [20] investigated the preferred digital marketing approaches among Gen Z college students in Davao City. They approached the problem by understanding how different digital attributes drive consumer interaction. Using a conjoint analysis approach through the PAPRIKA method, their study assessed attributes including social media, publicity types, and content types. They demonstrated that in the specific context of Davao City's Gen Z student market, content type (26.7%) and platform choice (24.7%) are the most influential factors. This proves that this demographic favors engaging, interactive content over traditional online advertisements [20]
4. *Consumer Decision-Making:* Gen Z consumers in Davao City often consider their experience and the physical ambiance of a coffee shop before they return. According to a study by Alcraz et al. [4], they modeled consumer behavior using the "Theory

of Planned Behavior” to assess Gen Z tourists’ attitudes toward local coffee shops. They found a correlation between experiences and visit intention (0.693) and a correlation of (0.614) between attitudes and visit intention [4].

2.2 Text Analysis of Consumer Reviews

Consumer reviews have become a primary source of public sentiment in digital markets, and processing them effectively requires both categorizing the type of feedback being expressed and understanding its structure [33] [24]. Lehto et al. [33] addressed this by analyzing unstructured consumer feedback from virtual travel agencies and categorizing reviews into three categories (complaints, compliments, and satisfaction). Classifying text along these polarities enables businesses to isolate specific problem areas and convert large volumes of feedback into actionable insights for managing consumer loyalty.

Structuring the text itself requires further decomposition. Herhausen et al. [24] noted that business researchers often lack coherent frameworks for processing digital stakeholder communication, and proposed applying a message design logic lens to break reviews into three components (background, situation, and intention). Organizing unstructured language into these elements provides researchers with a replicable method for translating consumer-generated text into measurable insights [24].

However, the explosion of user-generated content (UGC) on platforms like Google Maps has turned local market analysis into a massive ”Big Data” challenge. While consumer reviews provide valuable feedback, the huge volume of unstructured text renders manual analysis employing traditional qualitative methods impractical [60]. Almuayqil et al. (2022) [5] confronted the challenge of extracting reliable sentiment from large bodies of redundant and inconsistent review data. Pairing sentiment analysis algorithms with a preprocessing pipeline that included tokenization, lowercasing, and punctuation removal, they found that rigorous data normalization substantially improved a classifier’s ability to handle noisy text and detect sentiment polarity accurately [5].

Extracting meaningful insights from thousands of Davao City coffee and matcha shop reviews cannot rely solely on manual reading [30]. Natural language at scale also introduces problems of sparsity, dimensionality, and ambiguity that rule-based models and simple frequency counts are ill-equipped to handle [2]. Automated preprocessing pipelines paired with advanced sentiment analysis are, therefore, necessary to produce reliable, localized findings from this volume of consumer-generated [46] [36].

2.3 Code-Switching

Code-switching presents consistent uses in examining reviews from the Philippines. This refers to how writers or speakers switch between English, Tagalog, and other regional languages within the same text. [44] [15]. Mixed-language varieties such as Taglish (Tagalog-English) and Bislish (Bisaya-English) present particular difficulties for NLP models trained on monolingual English corpora [25]. A Davao-based review, for instance, might use the Bisaya term *lami* (meaning ”delicious” [63]) alongside English or Taga-

log commentary about service. A standard English lexicon would register no positive sentiment in such a sentence at all [44].

There are other issues where code-switching involves the simultaneous blending of distinct grammatical rules and syntactic structures, which standard computational models are not designed to handle [44] [62]. Patil et al. [46] noted that most sentiment analysis tools in big data contexts are built around monolingual English, leaving them poorly suited for low-resource languages, particularly when multiple languages appear within a single post. Conventional feature extraction methods tend to treat hybridized or region-specific expressions as out-of-vocabulary errors, discarding the semantic content that gives those expressions their meaning.

Other works on Philippine consumer data confirm this. Cosme and De Leon [15] discovered that lexicon-based tools perform poorly on Filipino-English reviews because single-language systems cannot reliably determine polarity when inputs shift between languages mid-sentence. Ang Ngo Ching et al. [8] reached a similar conclusion. Observing that English-only dictionaries lack the syntactic coverage necessary to process Taglish text without significant accuracy loss.

Through these limitations, techniques that can tolerate inconsistencies in grammar, vocabulary, and context across languages are essential. Maceda et al. [36] make this point in the context of code-mixed social data, where the input rarely conforms to the clean, monolingual text that traditional models assume. Multilingual transformers are suited to this environment, as their shared subword vocabularies and self-attention mechanisms enable fine-tuning on localized, code-mixed corpora, enabling sentiment classification across blended language boundaries [62] [36].

2.4 Lexicon-Based Approaches

Sentiment analysis has long relied on tools such as VADER, developed by Hutto and Gilbert in 2014 [27]. The model combines a curated lexicon with grammatical rules for negation and degree modifiers, and it was designed specifically for the messiness of social media: slang, emojis, and punctuation stacking. On standard English data, it performs well, achieving an F1-score of 0.96 [27]. That performance held in practice as Mathayomchan and Taecharungroj (2020) [39] used VADER to analyze 935,000 Google Maps reviews from 5,010 UK restaurants. Mapping lexical signals across food, service, atmosphere, and value, they found that food quality is the main driver of five-star ratings, and poor service accounts for most one and two-star reviews, which is clean and useful for restaurant owners.

The limitations of this approach become apparent, however, when the input deviates from standard English. Ang Ngo Ching et al. [8] examined this problem in the context of Philippine online reviews, where Filipino-English code-switching and code-mixing are pervasive features of written discourse rather than exceptional occurrences. The researchers evaluated six sentiment analysis tools, including VADER, against an annotated dataset of 1,000 reviews sourced from Shopee and Google Maps, assessing performance at both the sentence and review levels. VADER recorded a macro-F1 score of 0.30, repre-

sentencing a 35% decline from its performance on standard English text. This is due to the model’s English-only lexicon, which lacks the vocabulary and syntactic coverage necessary to process Taglish, the hybrid register in which Filipino speakers routinely alternate between Filipino and English [8]. The findings indicate that lexicon-based methods are insufficient as primary classification tools for multilingual, code-switched text in localized contexts such as the Philippines [8].

Furthermore, when compared to learning-based models, lexicon methods are particularly ineffective for highly localized data, such as consumer reviews in Davao City. Standard English lexicons completely miss positive sentiment when encountering unique regional syntax and Bisaya terms, such as *lami* (Bisaya adjective meaning delicious [63]). Ortega [44] addressed this linguistic barrier by developing a neural network-based sentiment analyzer trained specifically on Bisaya-language text. The study demonstrated that both Bisaya and mixed Tagalog-English syntax consistently avoid standard dictionary mappings, leading to severe misclassification of consumer opinions when processed with conventional lexicon-based tools [44]. These findings reinforce a conclusion consistent with prior literature. While lexicon-based algorithms such as VADER offer a useful baseline, their dependence on monolingual English dictionaries makes them inadequate for text characterized by regional code-switching. This limitation, specifically the failure of English lexicons to account for Bisaya and Taglish syntax, is the reason this study favors learning-based models over traditional approaches.

2.5 Learning-Based Approaches

2.5.1 Machine Learning

Where lexicon-based approaches fail against ambiguity and context dependency, machine learning methods offer greater adaptability. Pratmanto et al. [49] applied Naive Bayes classification to Indonesian e-commerce reviews scraped from the Google Play Store, focusing on user feedback for the Shopee platform. Before classification, the text data was preprocessed through tokenization, stop-word removal, and stemming, then partitioned into training and testing subsets. The model achieved 96.67% accuracy, indicating that probabilistic classifiers can handle large volumes of user-generated reviews without the need for manually constructed sentiment lexicons [49].

Pakpahan et al. [45] addressed the challenge of granular, aspect-level analysis by evaluating four machine learning algorithms, which are Naive Bayes, SVM, Random Forest, and K-Nearest Neighbors, on 20,000 Indonesian reviews from Shopee and Tokopedia. Word2Vec was used for feature extraction to capture semantic relationships between terms. Of the four models, SVM performed best, reaching 91% accuracy in aspect extraction and 90% in sentiment polarity classification [58].

Ulhaq and Suprayogi [58] took a similar comparative approach, applying SVM, Random Forest, and Multilayer Perceptron (MLP) architectures to over 10,000 preprocessed Tokopedia reviews vectorized using TF-IDF. The MLP model achieved the highest accuracy of 87.21%, narrowly surpassing SVM, indicating that neural classifiers offer a modest but consistent advantage over traditional methods on preprocessed consumer text

[58].

However, while traditional machine learning algorithms demonstrate high performance on standardized datasets, their reliance on static feature extraction methods (such as TF-IDF) limits their ability to capture deep sequential context and complex semantic relationships [46]. Since these traditional methods treat text as isolated tokens or static vectors, they lack the representational capacity required to process severe linguistic complexity and code-switching effectively [62]. Consequently, these traditional architectures are insufficient for handling the highly nuanced, code-switched syntax (Taglish and Bislish) present in localized Davao City reviews. This critical limitation in processing dynamic, mixed-language structures naturally necessitates the deployment of advanced learning-based approaches, such as Multilingual Transformers, which are explicitly designed to capture these complex contextual relationships [36].

2.5.2 Multilingual Transformers

Where lexicon-based and traditional machine learning methods fall short in handling linguistic complexity, the literature points toward deep learning as the more capable alternative. Multilingual transformer models, particularly mBERT and XLM-RoBERTa, have gained considerable traction in code-switched text analysis. Their effectiveness stems from transfer learning in both models, which are pre-trained on large multilingual corpora [62] [46], which allows them to recognize sentiment across mixed-language patterns such as Taglish and Bislish before being adapted to a specific classification task [36].

The core difficulty in processing code-switched text is that most NLP models are trained on monolingual data, leaving them poorly equipped for sentences that blend two or more languages within a single utterance. Winata et al. [62] examined this problem by evaluating mBERT and XLM-RoBERTa across several code-mixed datasets. Their findings confirmed that while both models carry strong cross-lingual capabilities through shared subword vocabularies and self-attention mechanisms, fine-tuning on task-specific, code-switched data produces substantial gains in classification performance, establishing these architectures as well-suited backbones for mixed-language tasks [62].

The necessity of localized fine-tuning was directly taken up in the Philippine context by Maceda et al. [36], who analyzed public sentiment on higher education policy using social media data rich in Filipino-English code-mixing. They adapted mBERT's pre-trained linguistic knowledge to the syntactic patterns of their Taglish corpus, ultimately achieving an F1-score of 81.14% and outperforming Support Vector Machine baselines by a considerable margin [36]. The result reinforced the practical value of domain-specific fine-tuning when the target language environment departs substantially from standard monolingual text.

Furthermore, the sheer scale and variability of social media data present obstacles. Patil et al. [46] assessed sentiment analysis frameworks against unstructured data from social media platforms, where textual noise, informal jargon, and inconsistent grammar routinely disrupt conventional feature extraction. Their comparative evaluation found that transformer-based models consistently outperformed traditional machine learning methods in accuracy and robustness across datasets with high linguistic variability [46].

Salve and Tubil [51] demonstrated that XLM-RoBERTa performs well on Southeast Asian language contexts, specifically in handling the kind of rapid language switching and non-standard syntax that appears in bilingual, low-resource settings. In the Philippines, consumer-generated text mixes Tagalog and English in which creates problems for most natural language processing models. Informal spellings, slang, and unstable sentence structures produce ambiguities that some models struggle to resolve. To address this, Salve and Tubil [51] fine-tuned XLM-RoBERTa Large on 10,000 Taglish reviews, using a custom SentencePiece tokenizer and explicit ambiguity tagging at both the token and sentence levels. The model outperformed both its Base variant and mBERT, reaching a testing precision of 97.62%, an F1-score of 95.77%, and an ROC-AUC of 97.50%. These results suggest that XLM-RoBERTa’s pre-training across more than 100 languages gives it a meaningful advantage when classifying mixed-language text, which is directly relevant to the code-switched review data collected for this study [51].

These considerations bear directly on the analytical demands of this study. Consumer reviews from Davao City coffee and matcha shops are written in Bisaya-English and Tagalog-English [3][29][20]. These varieties use region-specific vocabulary, such as *lami* (a Bisaya adjective meaning ”delicious” [63]), that standard lexicon-based tools are unable to interpret. Thus, while this research is theoretically grounded in the Aspect-Based Sentiment Analysis (ABSA) framework utilized by Maceda et al. [36]. The study will implement XLM-RoBERTa as the primary classification engine. By substituting the baseline mBERT architecture with XLM-RoBERTa, the study retains the proven ABSA methodology for localized Philippine texts while significantly optimizing the model’s ability to process the complex, informal code-switching of Davao City consumers, as supported by the recent performance benchmarks of Salve and Tubil [51].

2.6 Web Scraping

Web scraping refers to the automated extraction of unstructured data from websites and its conversion into machine-readable formats [57][12]. In sentiment analysis research, it enables large-scale retrieval of user-generated content, such as consumer reviews, that would otherwise be impractical to collect manually [12][18].

Digital data collection methods generally fall into three categories: manual gathering, direct API access, and custom HTML or DOM scraping. API access is preferred where available, as it returns clean, structured data. Mathayomchan and Taecharungroj [39], for instance, drew on Google Maps data to analyze customer experiences across 5,000 businesses, extracting over 935,000 reviews to measure attributes such as food quality, service, atmosphere, and value. However, it has become increasingly difficult to replicate through official channels [39].

Official APIs have grown substantially more restrictive. Chen et al. [12] documented this shift, comparing eight data collection methods and finding that while APIs are straightforward to use, their export limits and associated costs make them impractical for large-scale research. Custom scraping scripts, by contrast, were able to gather sufficient data volume without those constraints [12].

This broader restriction has been described as the "APIcalypse" by Trezza [57]. Examined how platforms banning or severely capping free API access have disrupted large-scale digital research, finding through a survey of digital researchers that API limitations have pushed much of the field toward web scraping as the primary means of maintaining data volume and research viability.

Dongo et al. [18] compared official APIs against custom HTML scraping for real-time social media data collection and found that scraping was considerably faster and more flexible for high-volume retrieval. Crucially, when the scraped text was properly normalized, the two methods produced equivalent analytical results, suggesting that data quality need not be sacrificed for scale [18].

In the context of this study, the standard Google Places API limits extraction to 5 reviews per business, far below the volume needed to fine-tune a deep learning model. Following the approaches documented by Chen et al. [12], Trezza [57], and Dongo et al. [18], this study employs custom HTML scraping to work around that ceiling. The textual noise inherent to scraped data will be addressed through normalization and the preprocessing pipeline described by Maceda et al. [36], ensuring the model receives sufficiently clean input for fine-tuning.

2.7 Synthetic Data Generation

Synthetic text generation creates artificial data using either code-generated data from algorithmic/mathematical models or Large Language Models (LLMs) trained on small seed datasets of real data [61][22]. To deal with the problem of data scarcity, researchers have increasingly looked to LLM driven generation as a workaround. Wang et al. [61] tackled the issue of language models being too dependent on human written instruction data, which is both expensive and limited in quantity and diversity. Their solution was the "Self-Instruct" framework, an iterative setup where a vanilla GPT-3 model was prompted to essentially bootstrap its own dataset by generating new instructions, identifying task types, and creating corresponding input-output instances. After filtering and using over 52,000 purely synthetic instances to fine-tune the model, they saw a 33% absolute performance improvement. That is a pretty significant result, and it effectively matched models trained on large private human datasets [61].

Scaling synthetic generation is one thing, but applying it to structured tasks like ABSA introduces its own set of problems. Monfared et al. [40] found that raw LLM augmentation often struggles to maintain label consistency, meaning the generated sentences frequently fail to preserve the specific aspect level annotations that were actually requested. Their proposed fix was an agentic data augmentation method that uses iterative generation and verification to produce higher quality synthetic training examples. Under a ReAct style architecture, a generator agent first extracted a style policy to produce candidate sentences, while a separate evaluator agent checked that each generated review actually contained the right aspect terms and correct sentiment polarities before it got added to the dataset. Compared to a matched-prompting baseline across three ABSA subtasks, the agentic approach clearly outperformed raw prompting in label preservation [40]. When combined with real data, it also consistently gave stronger performance

gains, particularly for models like T5-Base, which ended up performing at a level more comparable to its more heavily pretrained counterparts [40].

However, efficiency and scalability is considered costly [22]. Relying too heavily on synthetic text introduces what is often called "model collapse," and it is a real concern. Guo et al. [22] looked directly at this risk, asking whether training models on data generated by their predecessors might permanently degrade linguistic richness. They simulated recursive fine-tuning experiments across various natural language generation tasks and applied specialized metrics to track how text structure changed over successive iterations. What they found was a consistent and significant decline in linguistic diversity, specifically in lexical, syntactic, and semantic richness. Models trained excessively on synthetic data eventually converge to uniform behavior and lose the "tails" of the original distribution [22].

Compared to traditional manual annotation, synthetic data generation is the more scalable option for dealing with data scarcity [22] [40]. As Wang et al. [61] showed, synthetic generation can rapidly produce massive datasets from small seed samples. Still, to avoid the linguistic homogenization that Guo et al. [22] warned about, synthetic generation has to be carefully balanced with authentic data rather than used as a straight replacement.

To operationalize this balance and prevent generative model collapse, recent research has established exact mathematical thresholds for mixing authentic and synthetic data. He et al. [23] investigated this boundary within a recursive learning framework by evaluating customized weighted training schemes to optimally integrate real and synthetic batches. Their methodology spanned three distinct statistical scenarios—exact finite-sample calculations for Gaussian distribution estimation, asymptotic analysis for Generalized Linear Models (GLMs), and non-parametric discrete distribution approximations—which were subsequently validated using Monte Carlo simulations. The findings demonstrated that naive data mixing is strictly suboptimal. Crucially, they derived a unified analytical expression revealing that when mixing batches of equal size, the optimal weight assigned to real data is exactly the reciprocal of the golden ratio (approximately 61.8%). This dictates a theoretical ceiling, indicating that synthetic data influence must be kept below approximately 38.2% to prevent performance degradation.

With these findings in mind, this study will use the Self-Instruct framework from Wang et al. [61] to augment training datasets, though model collapse as discussed by Guo et al. [22] remains a real boundary. Following the safety limit defined by He et al. [23], this study limits synthetic data at 30%. Synthetic generation serves purely as a supplement, not a replacement for actual data collection. The pipeline generates synthetic Bislish and Taglish reviews only for aspect-sentiment pairs left underrepresented after manual annotation. Keeping this ratio well below the 38.2% threshold gives the model a balanced class distribution while preserving the distinct linguistic nuances of Davao City consumer reviews.

2.7.1 Semantic Validation and Diversity Filtering

While limiting the scope of synthetic generation mitigates macro-level model collapse [22], batch-generated instances still suffer from micro-level homogenization, where LLMs produce lexically varied but semantically identical text (e.g., swapping basic synonyms without altering core sentence structure)[52]. To enforce genuine syntactic variance, modern methodologies employ vector-based semantic filtering before dataset integration.

Converting generated text into sparse feature vectors via TF-IDF allows researchers to mathematically measure lexical overlap [43]. By computing the cosine similarity between the new synthetic vectors and the authentic seed dataset, we can discard any instances that cross a strict similarity threshold. This filtering acts as a built-in regularization step. It prevents dataset dilution and forces the downstream transformer to map actual linguistic boundaries, rather than just memorizing repetitive phrasing and spiraling into model collapse [65].

2.8 Aspect-Based Sentiment Analysis (ABSA)

General sentiment analysis gives a broad picture of consumer satisfaction, but it tends to fall short when reviews are complex, when a single sentence praises one thing and criticizes another in the same breath [13][17][7]. Aspect-Based Sentiment Analysis (ABSA) addresses this by pulling out specific entities or topics and identifying the sentiment attached to each one individually [13]. For a saturated market like the local coffee and matcha industry, where reviews frequently blend Bislish and Taglish to talk about distinct things like taste or service, having a robust ABSA methodology matters if the goal is actually actionable, granular insight.

One of the bigger challenges in applying ABSA to regional service sectors is code-switching, which tends to quietly demolish the accuracy of standard NLP tools [17]. Dionela et al. [17] looked at this problem specifically within Taglish, a low resource, code-switched blend of Tagalog and English that shows up constantly in Filipino consumer reviews. They built a pipeline comparing a traditional rule-based system, a generative LLM, and fine-tuned Gemma-3 models for identifying and extracting both explicit and implicit aspects. The generative LLM came out ahead across all tasks, which suggests that advanced generative models handle the messy syntactic structures and informal phrasing of Philippine code-switched reviews better than the alternatives [17].

Getting the aspects out is only part of the problem though. Accurately classifying the sentiment and emotional nuance inside code-mixed text is its own challenge [53]. Suciati and Budi [53] tackled this in the context of code-mixed Indonesian reviews, pointing out that standard ternary sentiment, positive, negative, neutral, often lacks the depth needed for nuanced consumer analysis. They compared traditional machine learning algorithms like Decision Trees, Random Forest, and Support Vector Machines against deep learning approaches like GRU and BiLSTM for multi-label classification across restaurant aspects such as food, price, service, and ambiance. The result was that traditional ensemble methods, specifically Random Forest and Extra Trees, actually outperformed the more complex deep learning models on certain tasks and achieved high F1 scores as

high as 89.54% for food and 92.65% for service. The takeaway is that for highly localized, code-mixed datasets, model architecture must be matched carefully to the specific aspect being evaluated rather than just defaulting to whatever is most complex [53].

Furthermore, capturing the contextual meaning of localized slang and code-switching also requires more than standard text representation [7]. Amalia and Winarko [7] identified this gap with traditional word embeddings, which struggle to track how the meaning of a word shifts depending on context, particularly in unstructured Indonesian restaurant reviews. Their fix was a hybrid architecture combining a Convolutional Neural Network (CNN) for feature extraction with various contextualized word embeddings, comparing BERT, ELMo, and Word2vec across aspect classification tasks. The ELMo-CNN combination came out on top, which reinforced the idea that pairing deep contextualized embeddings with a solid neural classifier meaningfully improves how well a model reads the nuanced, localized semantics of Southeast Asian consumer reviews [7].

With these findings as a guide, this study will adopt the multi-label aspect framework validated by Suciati and Budi [53] to analyze the kind of unstructured, fragmented feedback that shows up regularly in Davao City café reviews. An XLM-RoBERTa architecture will handle the multi-label classification, isolating and evaluating specific aspect-sentiment pairs. The goal is to make sure that the large volume of complex, code-mixed reviews actually gets converted into structured, usable insights that reflect the operational realities of local coffee and matcha businesses rather than just producing generic sentiment scores.

2.8.1 Aspect Extraction

To identify specific attributes or features from opinionated texts, an Aspect-based sentiment analysis (ABSA) is conducted, where aspect extraction plays a key role. However, unlike traditional sentiment analysis methods, rather than assigning the entire text a single polarity, ABSA identifies a review’s varying aspects along with their respective polarity. Since multiple sentiments can be expressed in a single review, ABSA provides a more detailed understanding of user feedback [41].

Mukherjee and Liu [41] conducted a study and identified the limitations of existing aspect extraction approaches, particularly those that either extract aspect terms without proper categorization or rely on unsupervised topic modeling, which may produce aspects that are not meaningful or aligned with user needs. Hence, aspect categorization is subjective, as different applications require different categories of the aspects. For that, they introduced a semi-supervised modeling approach for aspect extraction and categorization using user-given seed words. Their models, SAS, and ME-SAS also model sentiment jointly with extraction of aspect terms and grouping them into categories.

In contrast, Poria et al. [48] dealt with the problem of implicit aspects which are neglected by previous methods. They noted that implicit aspects are typically expressed through contextual clues rather than direct mentions, making them more difficult to detect. To this end, they proposed a rule-based approach, common-sense knowledge, and dependency parsing, which extract both explicit and implicit aspects. The implicit aspect clue is determined and mapped to an aspect category with linguistic rules and semantic

information. Their research work showed that the detection accuracy of the developed approach is higher than existing techniques.

Suryabrata et al. [55] addressed the challenges of aspect extraction in the case of multilingual and code-mixed scenarios by proposing a transformer-based ABSA framework relying on fine-tuned multilingual models, including mBERT and XLM-RoBERTa. Their major innovation is the sentence-pair classification strategy, which reformulates aspect extraction as a Natural Language Inference (NLI) task. The review sentences were then paired with candidate aspect-sentiment expressions in the framework, enhancing the capture of contextual relations between aspects and sentiments. For evaluation, their samples set of 20,000 aspect-level instances were split into (70%) training, (15%) validation, and (15%) test sets using random sampling without stratification. The sample splits were reused for all models to observe a fair comparison between single-sentence classifiers and their proposed sentence-paired models.

Their results showed that the XLM-RoBERTa achieved an accuracy of 97.20% and an F1-score of 0.9729, while mBERT showed more stability in performance across the validation sets.

Among the reviewed studies, the work of Suryabrata et al. [55] best fits the proposed framework. This, combined with a sentence-pair (NLI) approach, allows for more accurate identification of implicit and explicit aspect-sentiment pairs in a multilingual-code switched environment with a fine-tuned version of mBERT. The approach fits with the study objectives since the information to be analyzed is unstructured, in different languages, and context-specific consumer reviews.

2.8.2 Aspect Taxonomy

Building an aspect taxonomy is one of the more consequential early decisions in any ABSA study because it shapes what the model can and cannot detect downstream [10][9]. Older approaches relied on fixed dictionaries, which tend to miss necessary attributes and break down quickly when the text is informal or syntactically irregular [17]. More recent work has moved toward using Large Language Models to induce taxonomies directly from raw data, without requiring predefined categories or domain-specific seed sets. In a study

In a study, Brady and Islam [10] took this direction when analyzing political advertising content on social media in the 2024 U.S. presidential election. Since the content was highly subjective and shifted rapidly, they built an end-to-end framework that combined unsupervised clustering with prompt-based LLM inference. The model constructed a taxonomy iteratively, without needing seed sets or prior domain knowledge, and the resulting topic labels were more semantically coherent and interpretable than those produced by classical methods [10].

Boughanmi et al. [9] applied a similar approach to food and beverage feedback, working with 20,000 Yelp reviews of Starbucks locations. They tested eight prompt variants against manual human coding and found that the LLM consistently matched human annotators while processing each review. Beyond speed, the extracted attributes showed strong predictive validity for customer ratings, which suggests that generative extraction

is not just efficient but substantively accurate [9].

The challenge becomes considerably harder in low-resource and code-switched settings. Dionela et al. (2026) [17] examined this directly using Filipino e-commerce reviews written in Taglish, where standard NLP tools and rigid rule-based systems tend to fail. Their pipeline compared rule-based extraction, a fine-tuned Gemma-3 model, and a generative LLM (Gemini 2.0 Flash). The generative model outperformed the other approaches across all extraction tasks, handling implicit aspects and informal syntax without relying on any predefined lexicon [17].

This study follows the framework used by Dionela et al. [17] for its taxonomy discovery phase. Reviews from Davao City coffee and matcha shops are written heavily in Bislish and Taglish, which means a standard industry taxonomy would likely miss locally relevant concerns like specific ambiance traits or parking availability. Deploying a generative LLM as an unsupervised aspect allows the taxonomy to emerge from the actual language consumers use, rather than from categories decided in advance [17].

2.9 Active Learning Frameworks

Fine-tuning data demands significant amounts of high quality labeled data. When it comes with large volumes of scraped consumer reviews, exhaustive manual annotation quickly becomes a serious bottleneck [42][56][38]. Tharwat and Schenck [56] examined this problem, noting that even with an abundance of free, unlabeled text from digital platforms, the time and cost of manual labeling remain among the biggest obstacles to building supervised learning models. Their survey pointed to Active Learning as a practical workaround. Rather than labeling an entire dataset blindly, AL uses specific query strategies to find the most informative and uncertain data points in the instance space. Focusing human effort on high-value samples achieves strong model generalizability at a fraction of the usual labeling cost [56].

Considering that the feedback is multi-dimensional and complex. Muralidhar and Hulipalled [42] addressed the time and cost constraints associated with using expert human annotators for ABSA on unstructured social media data. They built a Deep Active Learning framework designed to predict aspect-based ratings for smartphones, using uncertainty sampling to iteratively pick out only the most confusing or informative reviews for human annotation, while a lexicon-based filter removed uninformative neutral statements at the same time. The DAL model hit 92% accuracy, which is a strong result, and it showed that intelligently querying uncertain samples can drastically cut down annotation effort without hurting predictive performance [42].

The same logic extends to fine-tuning modern language models. Margatina [38] explored this in her work on data efficient language models, starting from the observation that not all training examples contribute equally to the learning process even when large amounts are available. By simulating various active learning algorithms, she showed that AL guides training more intelligently by selecting only the most informative unlabeled data points at each iteration. Optimizing data selection this way consistently outperformed random sampling, letting language models reach strong performance with

far fewer human labeled instances [38].

These findings directly inform how this study will handle the annotation bottleneck for Davao City consumer reviews. Following the approach demonstrated by Muralidhar and Hulipalled [42], the system will use active learning to selectively surface only the most uncertain Bislish and Taglish samples for manual review. This keeps human effort manageable while ensuring the XLM-RoBERTa model is trained on the most informative data available, which is the central aim of bridging raw web-scraped text and a robust aspect-based classification system [53][17].

2.10 Manual Labeling in Sentiment Analysis

Active learning mechanisms are good at identifying the most uncertain or informative instances in a text corpus, but identifying them is only half the problem [34]. These algorithms cannot actually resolve the ambiguity on their own. What they need, and what every active learning pipeline fundamentally depends on, is a human annotator to provide the definitive ground truth for those flagged edge cases [34]. This is why manual labeling is not just a supplementary step but a necessary one in any sentiment analysis workflow [1][6].

In the Philippine context, where text in localized communications tends to be noisy and heavily context dependent, determining the situational validity of messages requires careful, extensive analysis. Ablazo [1] addressed this by using crowdsourcing through Google Forms, recruiting 250 undergraduate students from Ateneo de Davao University to serve as annotators. The dataset was split into batches, and each instance was independently labeled three times using predefined situational categories. A custom script then checked for annotator agreement, accepting a label only when at least two out of three annotators reached the same conclusion. From a total of 12,054 cleaned instances, the process produced 11,340 entries with a matched majority agreement. Instances where the annotators could not agree were discarded, so only labels that actually had consensus were passed on to the automated classifiers [1].

The question of how to direct that human effort more efficiently is what Li et al. [34] tried to answer. They note that deep learning models need large datasets, but relying on instance by instance manual labeling makes it hard to balance label quality with cost. Programmatic labeling can scale up, but it produces noisy results. Active learning, on the other hand, struggles with cold start problems when there is not enough labeled data to begin with. To address these limitations together, Li et al. [34] developed the DALL framework, which combines data programming, active learning, and large language models into one integrated labeling system. Rather than annotating random samples, DALL uses active learning strategies like uncertainty sampling, query by committee, and abstain sampling to flag the instances most likely to confuse or conflict with current model predictions. These are then routed to human annotators, who correct the labels and refine the rules [34].

Fine-tuning pretrained language models for Aspect-Based Sentiment Analysis is another difficulty since the annotation needs to be both granular and large in scale.

Alturayef et al. [6] point out a specific problem with standard active learning in this context. Algorithms often query instances that are semantically similar to each other, thus, human annotators end up labeling the same kinds of phrasings repeatedly without actually adding new information. To fix this, Alturayef et al. [6] proposed the EASE framework, which builds diversity and data augmentation into the active learning process. Their method calculates contextual similarity scores using Sentence BERT and averages those with model confidence scores, so that the samples flagged for human review are not only uncertain but also meaningfully different from what has already been labeled. After annotation, those edge cases go through Easy Data Augmentation to generate additional training samples synthetically. By filtering out redundant instances before they ever reached human annotators, the framework reduced the total number of manually labeled samples needed by somewhere between 88% and 94% across benchmark datasets [6].

Building on these findings, this study adopts the active learning and data augmentation principles from Alturayef et al. [6] to guide its manual annotation phase. The dataset is made up of highly contextual, code-switched Taglish and Bislish reviews from local coffee and matcha establishments, so exhaustive manual labeling is simply not practical given the scale and variety of the data. By applying the framework’s dual requirement of uncertainty and sample diversity, the pipeline makes sure human annotators are only asked to evaluate complex, non-redundant linguistic edge cases rather than running through similar reviews that do not add anything new [6].

2.10.1 Inter-Rater Reliability

Once the complex, non-redundant edge cases are identified for human review, keeping the resulting annotations consistent becomes its own challenge. Using a majority voting strategy, like the consensus approach Ablazo [1] demonstrated, provides a clear label for subjective and code-switched text, but relying only on simple percentage agreement is often not enough for machine learning applications that need more rigorous validation.

The importance of choosing the right IRR metric for qualitative annotations is a topic the literature has addressed fairly directly. El Dehaibi and MacDonald [19] looked at IRR measures for qualitative annotations in machine learning datasets and found that running an IRR check before training provides a necessary internal validity step. They explain that Cohen’s Kappa corrects for chance agreement between exactly two raters on categorical items, but once a study involves three or more raters, Fleiss’ Kappa becomes the appropriate metric.

Chamid et al. [11] show what Cohen’s Kappa looks like applied to a more demanding annotation scenario. Their study evaluated labeling consistency for ABSA using 4,307 Indonesian marketplace reviews, where two expert linguists independently labeled aspect detection, sentiment classification, and aspect class. The resulting Kappa values were 0.909 for aspect detection, 0.893 for sentiment classification, and 0.971 for aspect class, all falling in the almost perfect agreement range. These scores confirmed that the annotated dataset was reliable enough to actually train accurate classifiers on.

When annotation design scales up to three or more raters, Fleiss’ Kappa is discussed. Krušić [32] illustrates this in building a sentiment annotated corpus of historical

texts for NLP, where three semi-expert annotators categorized sentences into positive, negative, neutral, and mixed sentiment.

Fleiss’ Kappa across the three raters came out to 0.405, which falls in the fair to moderate range. A majority vote requiring at least two out of three annotators to agree was used to finalize labels, and the overall majority vote agreement reached 92.5%. These results were considered sufficient to establish the annotations as a reliable enough corpus for fine-tuning machine learning models.

Following this established standard for multi-rater setups, this study utilizes Fleiss’ Kappa (κ) to validate the ground-truth annotations provided by the peer annotators. The metric is formally defined as:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \quad (2.1)$$

Where the observed agreement $\bar{P}$ and the expected chance agreement $\bar{P}_e$ are derived from the distribution of ratings across subjects and categories:

$$\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i \quad \text{where} \quad P_i = \frac{1}{n(n-1)} \sum_{j=1}^k x_{ij}(x_{ij} - 1) \quad (2.2)$$

$$\bar{P}_e = \sum_{j=1}^k p_j^2 \quad \text{where} \quad p_j = \frac{1}{Nn} \sum_{i=1}^N x_{ij} \quad (2.3)$$

- N = total number of subjects or text instances
- n = number of raters per subject
- k = number of categorical choices (e.g., Positive, Negative, Neutral)
- x_{ij} = number of raters who assigned the i -th subject to the j -th category

2.11 Theoretical Frameworks

2.11.1 Multilingual Transformers

This method of Multilingual Transformers is grounded in the methodology of Maceda et al. [36]. Their study highlights the need for multilingual models to process code-mixed data, demonstrating that traditional approaches are insufficient for analyzing text that blends languages such as English and Filipino due to inconsistencies in grammar, vocabulary, structure, and context. By implementing a fine-tuned multilingual Bidirectional Encoder Representations from Transformers (mBERT), they were able to capture contextual relationships across multiple languages. The sequential process for this framework is depicted in Figure 1. This approach establishes that multilingual transformers are well-suited for handling linguistically diverse datasets, supporting their application in studies involving

mixed-language inputs. Similarly, Patil et al. [46] emphasized that transformer-based approaches consistently outperform traditional feature-based models when handling noisy text. Therefore, multilingual transformers are considered an optimal modeling framework for analyzing unstructured, code-mixed consumer reviews where linguistic variability is prevalent.

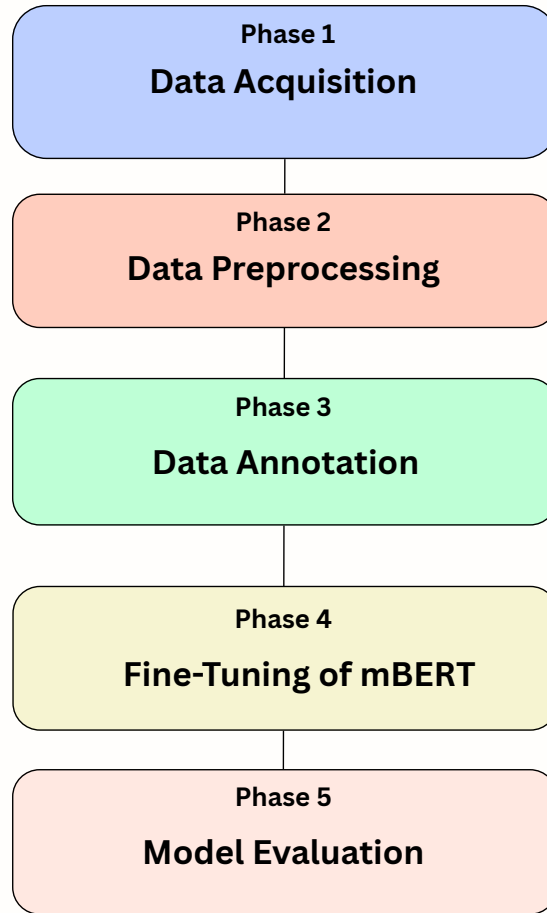


Figure 1: *Multilingual Transformers*

Phase 1: Data Acquisition The framework begins by collecting unstructured, code-mixed textual data from multiple social media platforms, such as Facebook, Twitter, and YouTube. Scraper libraries were utilized to systematically retrieve relevant posts, tweets, and comments using predefined keywords and sources [36].

Phase 2: Data Preprocessing The noisy and inconsistent nature of the code-mixed dataset necessitated preprocessing to clean and standardize the data. Duplicate entries were removed using URL-based identification, and irrelevant content was filtered manually. Minimal preprocessing techniques were applied, including the correction of informal spelling, normalization of special characters, removal of un-

necessary whitespace and punctuation, and conversion of emojis into their textual representations. This approach preserves contextual meaning while improving data consistency [36].

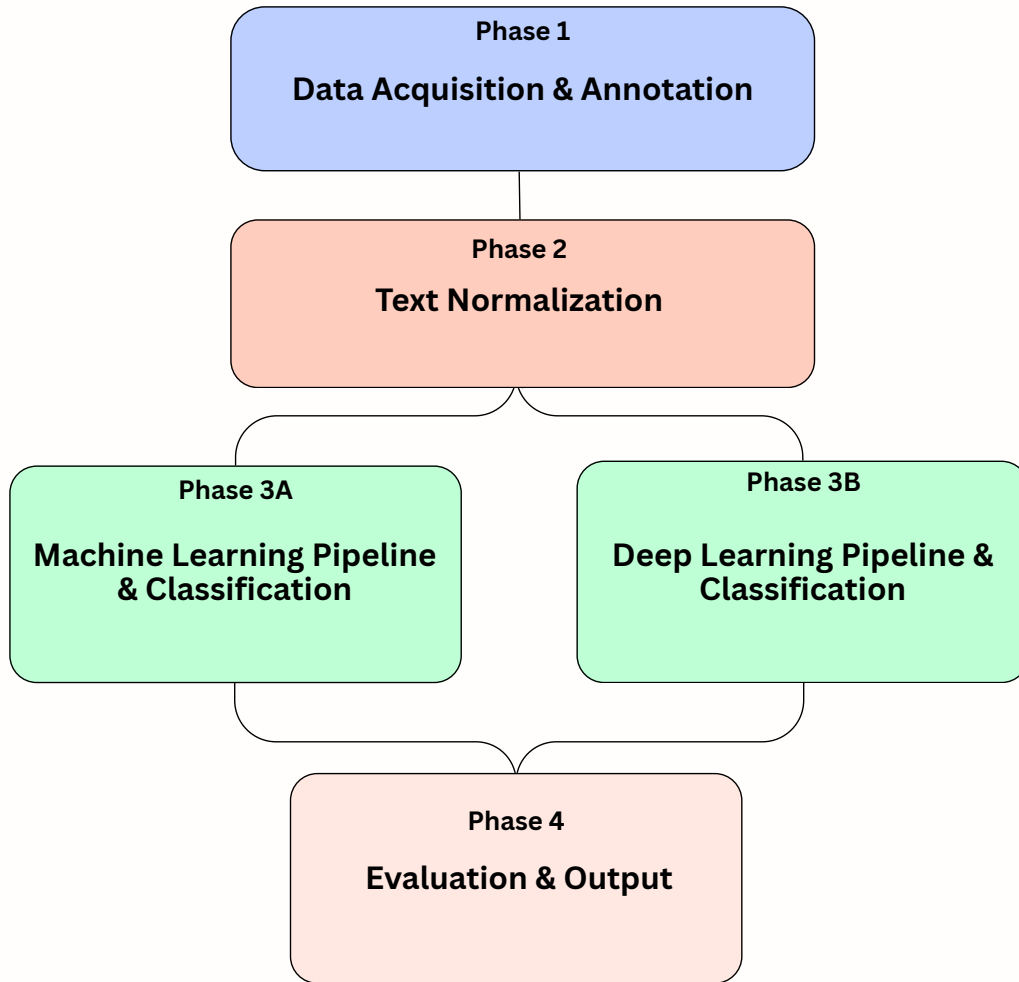
Phase 3: Data Annotation From the preprocessed dataset, data points were manually labeled and validated by experts. Each instance was classified into three sentiment categories: neutral, negative, and positive, encoded as 0, 1, and 2, respectively. The annotated dataset serves as the ground truth for training and evaluating the sentiment classification model, following established annotation practices [36].

Phase 4: Fine-Tuning of mBERT The framework utilizes a fine-tuned Multilingual Bidirectional Encoder Representations from Transformers (mBERT) model for sentiment classification. Using transfer learning, the model will leverage pre-trained global linguistic knowledge and adapt it to the specific variations in the mode-mixed dataset. Built on the Transformer architecture, mBERT captures contextual relationships across multiple languages, making it suitable for code-mixed data. The model processes input sequences using special tokens such as [CLS] for classification and [SEP] for segmentation, where the final hidden state of the [CLS] token is used for prediction. Fine-tuning was performed using task-specific parameters, including learning rates ($2e - 5$, $3e - 5$, $5e - 5$), batch sizes (16 and 32), and four training epochs. A maximum sequence length of 160 tokens was applied, and class weights were used to address class imbalance [36].

Phase 5: Model Evaluation The model's performance was evaluated using accuracy and F1-score metrics to assess its effectiveness in classifying sentiment in code-mixed text. The best-performing model configuration was then used to predict sentiment labels for the remaining dataset, enabling the analysis of overall sentiment distribution. This evaluation ensures the reliability and robustness of the framework in handling multilingual and unstructured data [36].

2.11.2 Aspect-based sentiment analysis

The method of Aspect-based sentiment analysis is grounded by the methodology by Su-ciati and Budi [53]. Their research addresses computational complexities of processing code-mixed (Indonesian-English) restaurant reviews by proposing a dual-pipeline classification architecture, as illustrated in Figure 2. Since standard single-language models fail to capture the nuances of bilingual reviews [25], their framework compares the potency of traditional Machine Learning (ML) models against advanced Deep Learning (DL) networks in performing multi-label classification for both sentiment and emotion.

Figure 2: *Aspect-based sentiment analysis*

Phase 1: Data Acquisition and Annotation The framework is based on collecting unstructured review data that contains naturally mixed formal language and informal code-switching. The data then goes through the annotation processes using crowdsourcing. Multiple annotators label the text based on predefined guidelines, and when annotators reach a disagreement on how something should be labeled, a majority vote settles the call [53].

Phase 2: Text Normalization (Pre-processing) As code-mixed social media text is highly noisy and inconsistent, the framework implements an extensive eight-step preprocessing layer. Abbreviations are corrected using self-developed bilingual dictionaries, while word stemming is handled by two dedicated tools: the Snowball Stemmer for English and the Sastrawi Stemmer for Indonesian. Emoticons are also converted into their text equivalents to ensure that meaningful emotional cues are preserved throughout the cleaning process [53].

Phase 3: Dual Classification Pipelines

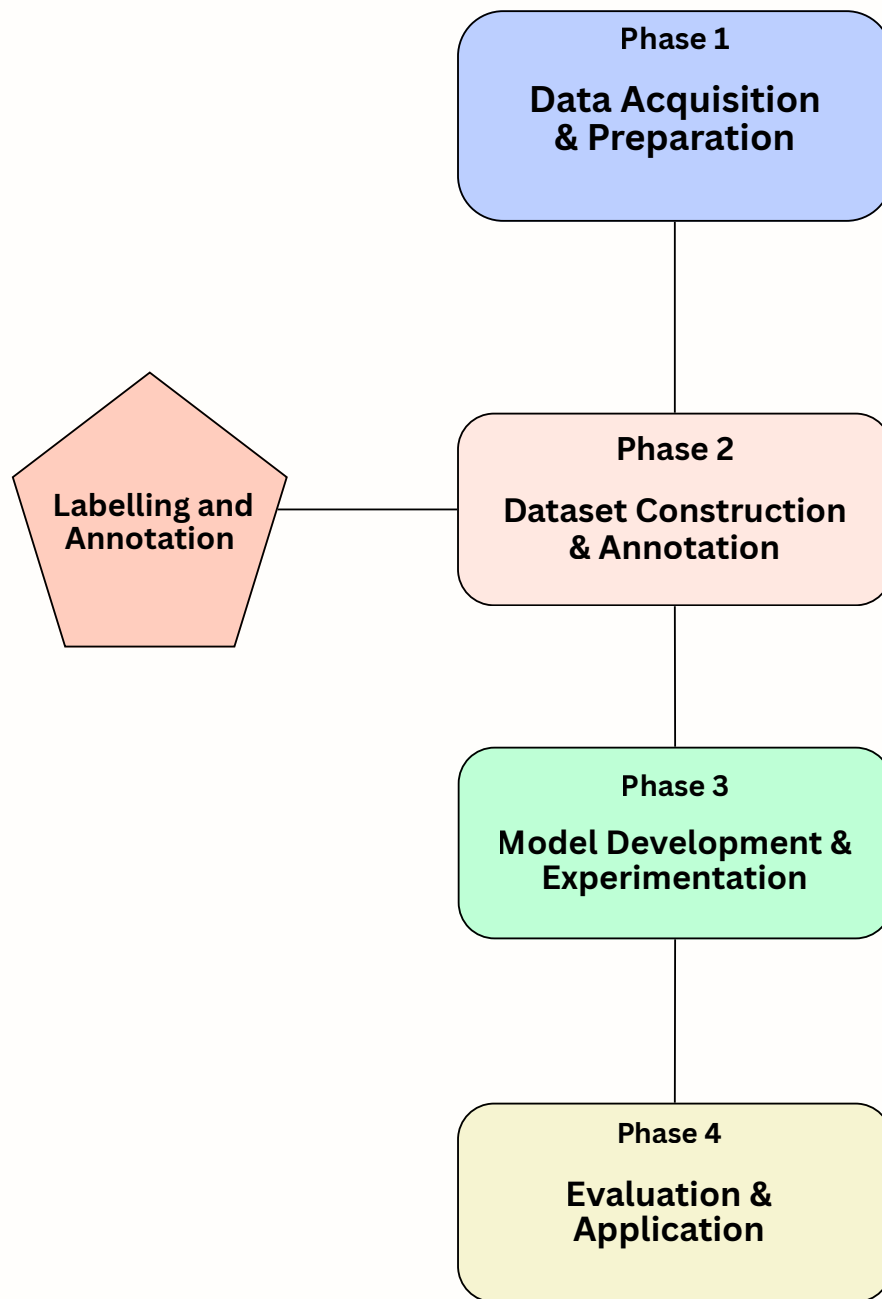
- *Machine Learning Pipeline:* The Machine Learning pipeline relies on traditional feature extraction methods through N-gram language models, such as unigrams, bigrams, and a combination of both, paired with Chi-Square for feature selection. Since aspect-based classification often involves reviews that address multiple targets at the same time, the framework applies problem transformation methods such as Binary Relevance, Classifier Chain, and Label Powerset to adapt standard algorithms including SVM, Random Forest, Decision Tree, and Extra Tree into multi-label classifiers [53].
- *Deep Learning Pipeline:* The deep learning pipeline takes a different approach by eliminating the need for manual feature engineering. Instead, it constructs custom word embeddings using the Skip-gram method to capture the semantic relationships present in code-mixed text. Deep learning architectures, specifically BiLSTM and GRU, are then deployed using a sigmoid activation function and binary cross-entropy loss to directly predict multiple aspect-sentiment pairs at the same time [53].

Phase 4: Evaluation and Output The performance of both pipelines is thoroughly tested using a 10-fold cross-validation technique. The framework evaluates each model based on its F1-Score, measuring how accurately it was able to isolate and classify sentiment and emotion across four distinct business dimensions: food, price, service, and ambiance [53].

2.11.3 Aspect Extraction

This framework draws from Natural Language Processing, transfer learning, and Aspect-Based Sentiment Analysis. The core problem with traditional sentiment analysis is that it reduces an entire review to a single polarity label, and that is simply too blunt for user-generated content [55]. In the study, hotel guests routinely praise one aspect of their stay while criticizing another in the same sentence. Recovering that granularity requires Aspect Extraction. It is the process of identifying which specific service attributes a reviewer is actually referring to [55].

The study uses mBERT and XLM-RoBERTa, two multilingual transformer architectures, to handle the code-mixed language that commonly appears in reviews from multilingual settings. Since no locally annotated dataset exists for this domain, zero-shot learning was adopted. The ABSA task is reformulated as a Natural Language Inference problem, which lets the models assess sentiment without needing labeled training examples from the target domain [55]. Figure 3 outlines the overall workflow and the structural phases of the study.

Figure 3: *Aspect Extraction*

Phase 1: Data Acquisition and Preparation Any NLP pipeline depends on the quality and representativeness of its corpus. For this study, over 100,000 reviews were collected from TripAdvisor and Google Reviews across 43 five-star hotels in Jakarta. The challenge at this stage is that user-generated content is unstructured, inconsistent, and often multilingual, with some reviewers switching between languages mid-text [55].

- *Collection and Preprocessing* Covers data collection and preprocessing. Duplicate entries and empty strings are removed to reduce model bias and prevent overfitting. A language filter then isolates reviews in Indonesian, English, and code-mixed text. This filtering decision was deliberate: mBERT and XLM-RoBERTa were pre-trained on multilingual corpora, and keeping the input data within those language boundaries ensures the models are operating within their intended distribution [55].

Phase 2: Dataset Construction and Annotation This is where Zero-Shot Learning and Aspect Extraction are present. A major hurdle in ABSA is the "cold-start" problem: the lack of massive, human-annotated datasets for specific domains (like Indonesian hotel reviews). This phase leverages the theory of Zero-Shot Transfer Learning, in which a model pre-trained on general tasks is used to classify unseen data without requiring domain-specific fine-tuning. Crucially, it relies on aspect extraction as the prerequisite mechanism to break down monolithic blocks of review text into granular, analyzable targets before any sentiment can be measured [55].

- *Zero-Shot Labeling and Aspect Extraction:* multilingual zero-shot classifier acts as a proxy annotator. The indispensable first step is aspect extraction, where the model automatically scans the raw text to identify what the user is talking about, categorizing the text into seven predefined aspects (Cleanliness, Facilities, Food, Location, Price, Room, and Service). Because a single review might discuss multiple aspects simultaneously. In the dataset, sentiments were isolates each extracted review-aspect pair into its own unique instance (generating 352,720 records). Only after these target aspects are successfully extracted and separated can a subsequent zero-shot pass accurately determine the specific sentiment (Positive, Negative, Neutral) attached to them [55].
- *Sampling and Splitting:* To optimize computational efficiency while preserving statistical significance, a representative sample of 20,000 aspect-level instances is extracted and rigorously partitioned (70% Train, 15% Validation, 15% Test) to ensure the model can be evaluated on strictly unseen data [55].

Phase 3: Model Development and Experimentation This phase is where the core computational work happens. Rather than treating ABSA as a standard classification task, the study reformulates it as a Natural Language Inference problem [55].

- *Sentence-Pair Construction and Tokenization:* Each review is restructured into a sentence pair. The review text serves as the premise, and a constructed auxiliary sentence. The model then predicts whether the premise entails the hypothesis. Framing the task this way gives the transformer explicit structural context about the aspect and sentiment orientation being evaluated. Tokenization is handled by subword algorithms appropriate to each model. WordPiece for mBERT and SentencePiece for XLM-RoBERTa, with the other being better equipped for the morphological complexity and code-switching present in the corpus, morphological variation and code-switching [55].
- *Training and Fine-Tuning:* One persistent problem with review datasets is se-

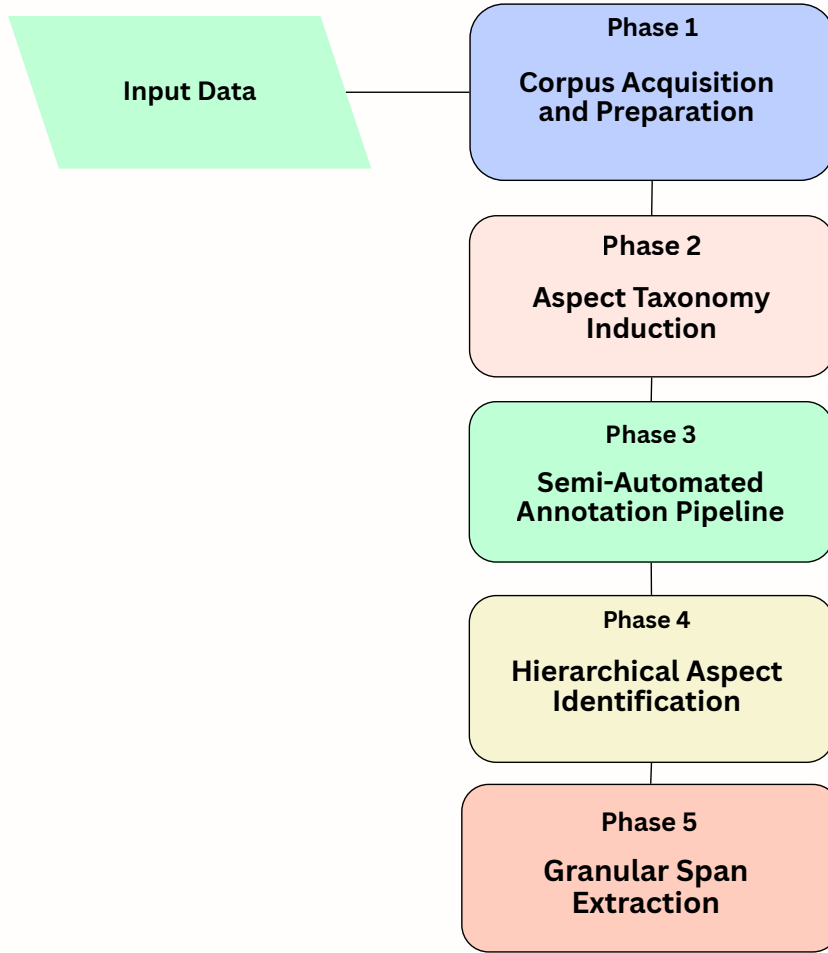
vere class imbalance, as positive reviews typically account for more than 90% of the data. A model trained on distribution can achieve deceptively high accuracy by defaulting to positive predictions while failing almost entirely on negative and neutral cases. Rather than correcting this through oversampling, which carries a real risk of overfitting, the framework applies Weighted Loss Functions under a cost-sensitive learning approach. Misclassifications of minority class instances are penalized more heavily during training, pushing the model to learn the distinguishing patterns of negative and neutral reviews rather than treating them as statistical noise [55].

Phase 4: Evaluation and application The final phase connects model outputs to practical use within a hotel management context.

- *Evaluation:* On imbalanced datasets, standard accuracy is an unreliable measure of model performance. A classifier that predicts positive sentiment for every review would exceed 90% accuracy on this corpus while failing entirely to detect negative or neutral cases. For this reason, the framework uses the F1-score as its primary evaluation metric, which computes the harmonic mean of Precision and Recall. Precision measures how many of the model’s predicted positives are correct, while Recall measures how many of the model’s predictions are correct. Confusion matrices and error analyses supplement these scores by identifying which linguistic structures consistently cause misclassification, providing diagnostic depth beyond what aggregate metrics alone can offer [55].
- *Application and Visualization:* Deploys the better performing architecture, XLM-RoBERTa, across the approximately 76,000 reviews that were not used in training or evaluation. The resulting sentiment predictions are mapped into visualizations including heatmaps and correlation charts. This translation step is what makes the model outputs usable in practice so that hotel management can identify which service aspects are consistently underperforming across reviews without having to interpret raw model inferences directly [55].

2.11.4 Aspect Taxonomy

This research builds on two related ideas: the Principle of Cross Lingual Semantic Resilience and the Theory of Generative Attribute Induction. Dionela et al. [17] incorporated both into their Aspect-Based Sentiment Analysis framework [17]. The claim is that generative language models can process messy, code-switched consumer feedback without relying on predefined word lists. Dionela et al. [17] points out that traditional Natural Language Processing fails in low-resource, mixed-language environments [17]. Their study of Philippine e-commerce reviews shows the problem clearly. Rule based models and small fine tuned models struggle with code-switched Taglish since they depend on keyword matching [17]. When consumers use informal language or imply an aspect without stating it directly, the system misses it. For instance, "It took a week to arrive" signals a delivery problem, but the keywords might not be obvious [17]. Refer to Figure 4.

Figure 4: *Aspect Taxonomy*

Phase 1: Corpus Acquisition and Preparation. Raw, unstructured code-switched consumer reviews are processed to create a baseline corpus [17]. The dataset contains 10,510 Taglish reviews from Google Maps and Shopee Philippines. Human annotators labeled both general and specific aspect categories [17].

Phase 2: Aspect Taxonomy Induction Rather than starting with predefined categories, this phase builds a Hierarchical Aspect Framework by combining four approaches: Open Coding, Latent Dirichlet Allocation for topic modeling, BERTopic for multilingual embeddings, and Generative LLMs to integrate the results [17]. The outcome was a taxonomy with 4 General Categories (Product, Delivery, Price, Service) and 21 Specific Aspects derived from the actual corpus [17].

Phase 3: Semi-Automated Annotation Pipeline To annotate implicit aspects at scale, human researchers first labeled 67 reviews as a reference set, achieving a Fleiss Kappa of 0.691. This reference set was then used as few-shot examples for Gemini 2.0 Flash, which automatically annotated the remaining reviews to create a

training dataset [17].

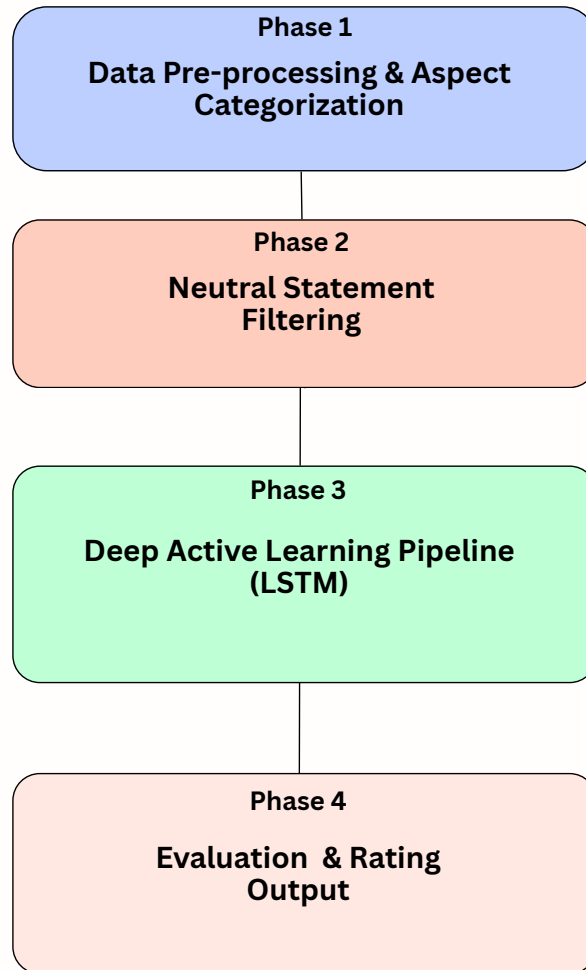
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Phase 4: Hierarchical Aspect Identification The framework uses conditional classification: broad domains like Product or Service are identified first, and these results determine which specific aspect classifiers run next. It was then compared to four approaches. A Rule Based system, Gemini 2.0 Flash, and two versions of Fine Tuned Gemma 3 (1B) [17]. The Generative LLM achieved a Macro F1 of 0.91, outperforming the others. The smaller fine tuned models had difficulty with noisy syntax and imbalanced classes in code-switched text [17].

Phase 5: Granular Span Extraction The system extracts exact text spans for the identified aspects, mapping code-switched vocabulary to the taxonomy. LLMs extracted numerical concepts like price accurately, but struggled with Over Extraction and Partial Matches when faced with variations in Taglish Product and Service terminology [17].

2.11.5 Active Learning

This method of Active Learning is grounded in the methodology established by Muralidhar and Hulipalled [42]. Their research addresses the high cost and time needed to manually label large social media datasets by proposing a Deep Active Learning pipeline as depicted in Figure 5 To reduce misclassification of objective and neutral statements, their framework combines lexicon-based techniques with human judgment to distinguish positive from negative sentiments. Text normalization, aspect categorization, and a Long Short-Term Memory (LSTM) network form the core of this pipeline. The model queries human experts only when classification confidence falls below a set threshold, resulting in efficient, aspect-specific sentiment predictions for mobile application reviews [42].

Figure 5: *Active Learning*

Phase 1: Data Pre-processing and Aspect Categorization Raw social media text presents several preprocessing challenges, such as misspellings, informal jargon, and hashtags that merge multiple words together. The framework addresses these using Ekphrasis, a tool that segments compound tokens, corrects spelling errors, and applies lemmatization to reduce words to their root forms without losing contextual meaning. Emojis and emoticons are converted to their text equivalents before the cleaned text is split into smaller phrases along punctuation and conjunction boundaries. Each phrase is then assigned to a product aspect such as "battery" or "camera" using predefined keyword dictionaries [42].

Phase 2: Neutral Statement Filtering Neutral, non-opinionated statements may distort classifier training if left in the dataset. To filter these out, the framework scores each text entry using a composite lexicon built from VADER, AFINN, and SentiWordNet. Since domain-specific language can complicate automated scoring "heavy" carries negative sentiment in a phone review but registers as neutral in a general lexicon. Then, a human annotator reviews the 100 most frequent words

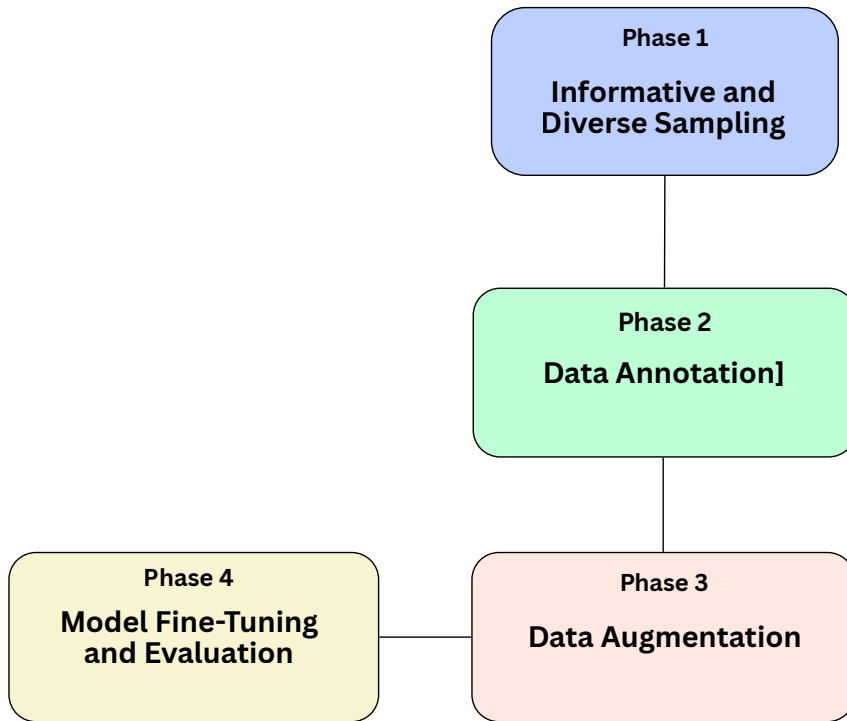
flagged as neutral. Words identified as false neutrals are reassigned sentiment polarities and reintegrated into the lexicon. The dataset is then recorded, and remaining neutral entries are discarded, leaving a training set composed entirely of opinionated samples [42]. While Muralidhar and Hulipalled [42] utilized standard lexicons for this filtering phase, this study’s conceptual framework adapts this approach to accommodate localized code-switching.

Phase 3: Deep Active Learning Pipeline The framework of the specified study uses an LSTM-based Active Learning strategy that selectively involves human input. A human annotator manually labels an initial batch of 5,000 reviews to train the model. The LSTM then processes the next 5,000 reviews: predictions made with at least 75% confidence are accepted automatically, while those falling below this threshold are routed back to the annotator for manual labeling. Each newly labeled sample is used to retrain the model, progressively improving its classification boundaries while keeping total annotation effort low [42].

Phase 4: Evaluation and Rating Output Model performance is evaluated against standard baselines, including SVM, using precision, recall, and F-measure. Beyond classification accuracy, the framework converts binary sentiment predictions into a 1-to-10 aspect-level rating scale, which is calculated by dividing the count of positive reviews by the total number of reviews for each aspect, thus producing a metric directly usable in commercial product evaluation [42].

2.11.6 Manual Labeling in Sentiment Analysis

This method of optimizing data selection for Aspect Based Sentiment Analysis (ABSA) draws from the methodology of Alturayef et al. [6]. Their study identifies manual annotation as a serious bottleneck when fine tuning pretrained language models for granular sentiment tasks. To address this, they proposed the Enhanced Active learning framework for Aspect-based Sentiment Analysis (EASE), demonstrating that filtering redundant data before it reaches human annotators can significantly reduce labeling costs without harming model accuracy [6]. The framework works by combining uncertainty sampling, sample diversity, and targeted data augmentation so that every manually labeled instance actually contributes something useful. The full sequential process is illustrated in Figure 6.

Figure 6: *Manual Labeling in Sentiment Analysis*

Phase 1: Informative and Diverse Sampling The framework starts by evaluating a large pool of unlabeled text. Rather than picking instances at random for human review, the algorithm computes two specific metrics to find the most valuable data points. The first is model uncertainty, which surfaces sentences the model finds ambiguous. The second is a contextual similarity score computed using Sentence-BERT, which checks whether a candidate sample is meaningfully different from the selected sample. Averaging these two scores together, the framework flags only the instances that are both uncertain and linguistically distinct from one another [6].

Phase 2: Data Annotation The informative and diverse instances identified in Phase 1 are then sent to human annotators for manual labeling. Since the algorithm removed semantically similar reviews, annotators spend their time on genuinely unique and complex patterns rather than variations of the same thing. This targeted approach reduces the total number of samples that need manual labeling by up to 94% compared to random sampling [6].

Phase 3: Data Augmentation To get more out of the newly annotated data and to handle potential class imbalances, the labeled instances go through Easy Data Augmentation (EDA). This step generates additional training samples from the seed data, which expands the overall training corpus while preserving the contextual integrity of the edge case reviews that were originally selected [6].

Phase 4: Model Fine-Tuning and Evaluation The final dataset which consists of a diverse manually annotated core supplemented by the augmented synthetic samples, is used to fine-tune the pretrained language model for ABSA. The model is then tested on benchmark datasets to confirm that classification accuracy holds up even with a drastically reduced annotation effort [6].

2.11.7 Synthetic Data Generation

To address data scarcity and class imbalance, the introduced study adopts the Self-Instruct framework developed by Wang et al [61]. The procedural steps of which are shown in Figure 7. The approach uses a pre-trained Large Language Model to generate training data from a small set of human-authored seed examples, expanding the dataset substantially without additional manual annotation.

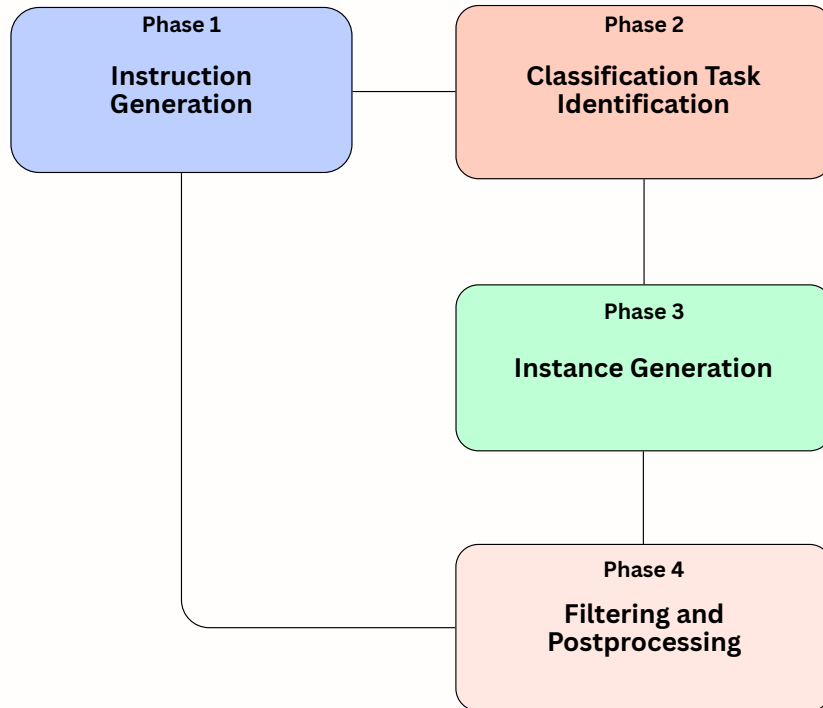


Figure 7: *Synthetic Data Generation*

Phase 1: Instruction Generation The process starts with a Task Pool containing a

small set of human-written seed examples. The LLM draws from these to generate new task instructions recursively, using existing examples as context for each subsequent generation [61].

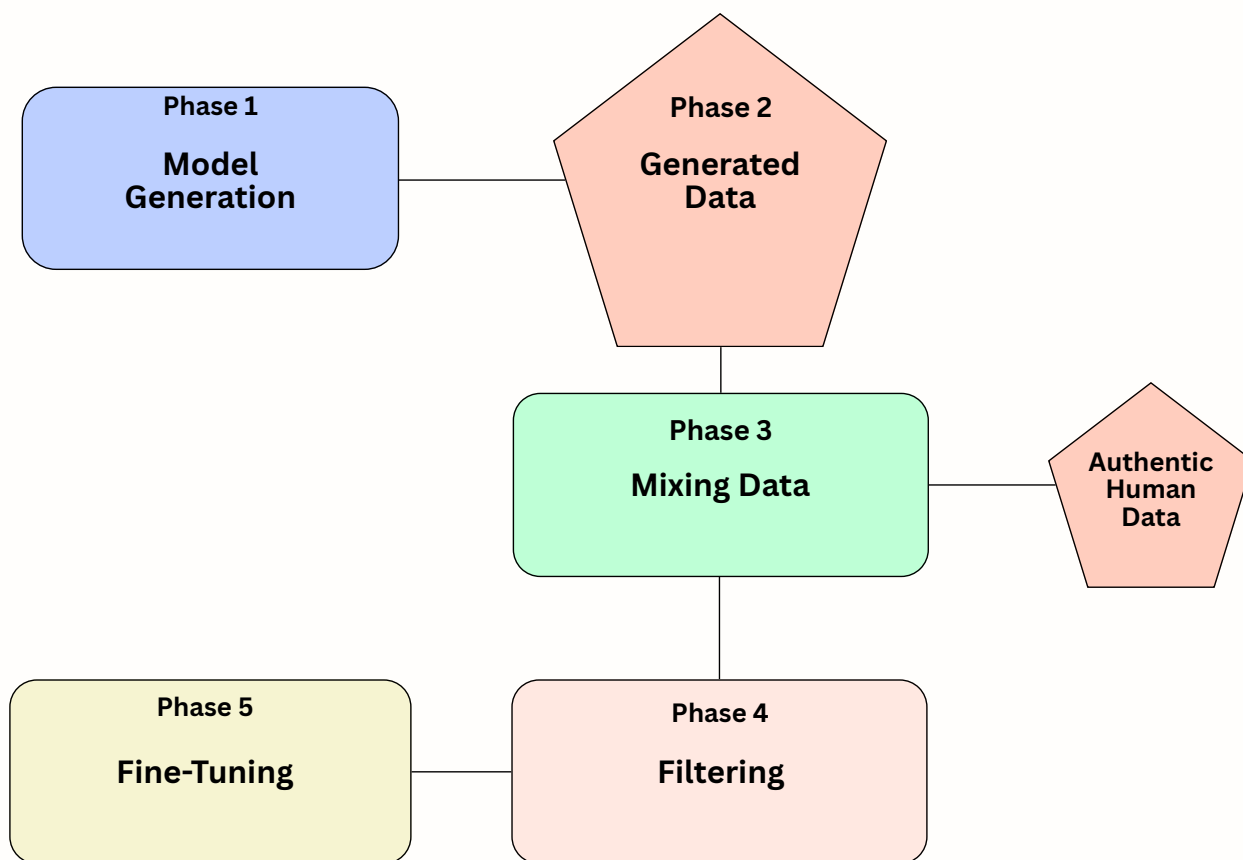
Phase 2: Classification Task Identification The model determines whether it is a classification task, such as assigning sentiment polarity to a review, or a non-classification task. This distinction controls how the input and output are structured in the next phase [61].

Phase 3: Instance Generation Using the instruction and its identified task type, the model generates a synthetic input-output pair. For classification tasks, the input is a synthetic text review, and the output is a set of aspect and sentiment labels. The structure differs for non-classification tasks based on what the instruction specifies [61].

Phase 4: Filtering and Postprocessing Before entering the Task Pool, generated instructions are checked for redundancy and validity. Instructions with a ROUGE-L overlap of 0.7 or above against any existing pool entry are discarded, including entries the system cannot parse. The remaining instances are added to the pool and become available as in-context examples for future rounds, growing the dataset over successive iterations [61].

2.11.8 Constraint of Synthetic Data: Model Collapse and Stabilization Framework

While the Self-Instruct framework by Wang et al. [61] establishes the methodology for generating synthetic data, scaling the dataset effectively requires managing the inherent properties of AI-generated text [22]. Preserving the code-switched syntax of localized dialect reviews shapes how the augmentation strategy must be designed. Figure 8 visualizes the framework for mixing synthetic and authentic data to mitigate these constraints. Thus, the theoretical framework of Guo et al. [22] demonstrates that training models on synthetic outputs causes a persistent decrease in linguistic diversity, a phenomenon that cannot be fully cured even when mixed with authentic human data. To resolve this, He et al. [23] introduces a Weighting-Based Stabilization Framework to calculate exact mathematical limits for synthetic integration.

Figure 8: *Synthetic Data Causing Model Collapse*

Phases 1 and 2: Generation and Diversity Loss The cycle begins with a language model generating a batch of synthetic text. Guo et al. [22] find that synthetic data consistently underrepresents the lexical, syntactic, and semantic range of human-authored text. This deficit compounds across generation rounds, producing diversity loss even in tightly constrained tasks.

Phase 3: Data Mixing To reflect a realistic training scenario, the framework introduces a mixing phase in which synthetic outputs are combined with a fixed proportion of human-authored data [22].

Phases 4 and 5: Persistent Degradation After filtering and fine-tuning the subsequent model, Guo et al. [22] show that the added human data produces only marginal gains in diversity. The core finding is that model collapse is not a problem mixing alone can solve. Diversity loss accumulates across iterations. Hence, reliance on synthetic data still causes a persistent, cumulative decline in linguistic diversity, eventually leading to *“model collapse”* [22].

Guo et al. [22] showed that while mixing human and synthetic data mitigates diversity loss, it does not completely eliminate it. Addressing this degradation, He et al. [23] developed a framework for recursive learning stabilization. They determined the optimal weight for real data sits at the reciprocal of the golden ratio (roughly 61.8%). This creates a hard ceiling: synthetic data must stay below 38.2

These mathematical limits directly shape the study’s augmentation strategy. Local coffee and matcha shop reviews in Davao City rely on code-switched Bislish and Taglish linguistic nuances that are notoriously difficult to replicate. Flooding the corpus with synthetic data would wash out the exact syntactic patterns the model needs to learn. Instead, synthetic generation via Wang et al.’s [61] Self-Instruct framework acts purely as a safety net. The pipeline only triggers when a specific aspect-sentiment class lacks enough real samples to be usable. Capping synthetic data at 30% keeps the project well within the 38.2% safety limit, balancing the training distribution without risking generative collapse.

To effectively analyze code-switched consumer reviews, this research integrates the Multilingual Transformers framework with the selective querying principles of Deep Active Learning, the unsupervised discovery of an Aspect Taxonomy, and the structural granularity of Aspect Extraction. Traditional feature-based pipelines relying on N-grams and algorithms like SVM will be excluded, as the literature consistently demonstrates that transformer-based architectures outperform them when processing noisy, unstructured text [46] [36] [62]. While the foundational Aspect-Based Sentiment Analysis (ABSA) framework established by Maceda et al. [36] utilized mBERT, this study implements XLM-RoBERTa as the primary classification engine, given its superior capacity to capture complex contextual relationships and resolve ambiguities in highly code-switched Southeast Asian languages [51].

Crucially, since traditional sentiment analysis reduces an entire review to a single polarity label, it is often too blunt for complex user-generated content where consumers frequently praise one attribute while criticizing another in the same sentence [55]. To address this, the framework first incorporates generative Aspect Taxonomy Induction [17] to discover localized operational dimensions from the raw text. This serves as a prerequisite phase for Aspect Extraction. Thus, ensuring the review text is broken down into granular, analyzable targets governed by a data-driven master dictionary rather than assumed categories. By identifying these specific service attributes and reformulating the classification task as a Natural Language Inference problem [55], the model can isolate and evaluate individual review-aspect pairs efficiently.

However, fine-tuning the classification engine requires a large volume of ground truth labels, and getting those labels manually is where the process tends to slow down considerably. To work around this, the methodology draws on the principles of the Enhanced Active learning framework for Aspect based Sentiment Analysis [6]. Instead of labeling everything or pulling samples at random, the pipeline uses uncertainty and sample diversity metrics to identify only the reviews that are genuinely ambiguous and linguistically distinct from one another. By narrowing the annotation queue down to these complex, non redundant cases, human effort gets directed toward the instances that actually matter rather than being spread thin across data that would not meaningfully

improve the model [6].

The framework uses the Self-Instruct methodology [61] to address data scarcity without triggering model collapse. Since mixing authentic data does not completely reverse the drop in linguistic diversity caused by relying heavily on AI text [22], the pipeline avoids bulk synthetic generation entirely. Instead, it applies He et al.’s [23] Weighting-Based Stabilization Framework. Synthetic data acts strictly as a targeted fallback capped at 30% of the training volume. This hard limit keeps the system safely below the 38.2% threshold of generative degradation.

Taken together, these approaches form a comprehensive theoretical framework that integrates XLM-RoBERTa’s robust classification capabilities, generative Aspect Taxonomy discovery, Active Learning driven by Fleiss’ Kappa validation, and mathematically constrained synthetic augmentation. The literature supports this specific combination as a highly effective architecture to balance class distribution, minimize human annotation costs, and mitigate generative collapse, all while preserving the complex, code-switched syntax that makes localized Bislish and Taglish reviews worth studying.

Chapter 3

Conceptual Framework

The study’s conceptual framework will be described using the Input-Process-Output (IPO) model, as illustrated in Figure 9, which maps the transformation of unstructured, multilingual consumer reviews into actionable market intelligence. The IPO model illustrated the flow of data from raw collection to final classification. Emphasizing the combined application of web scraping, active learning, synthetic data augmentation, and transformer-based sentiment analysis, thus, effectively addressing the limitations of traditional review evaluation in localized, code-switched domains.



Figure 9: *IPO Chart (Input-Process-Output)*

3.1 Input Phase

This defines the foundation for the data requirement. The primary data source consists of unstructured consumer reviews of local coffee and matcha businesses in Davao City. These reviews are gathered through scraping from Google Maps reviews. The collected raw reviews are then stored in segregated CSV files for coffee and matcha shops, prior to entering the processing pipeline. Its linguistic complexity is one of its unique characteristics. It is multilingual and features heavy code-switching among English, Tagalog (Taglish), and Bisaya (Bislish) [44].

3.2 Process Phase

This outlines the machine learning and data engineering pipeline for handling the input data.

1. **Custom HTML Scraping:** To gather enough data, the study bypasses the strict 5-instance limit of the standard Google Places API [21]. Researchers instead collect Google Maps reviews using a custom JavaScript scraper [59]. The workflow is semi-automated. Researchers manually navigate to a target establishment and scroll through its review section before triggering the script. The utility then extracts star ratings and text directly from the loaded DOM, filtering duplicates and handling basic text cleaning on the fly. The final records are exported into CSV files, which are split as needed to streamline downstream preprocessing.

A sufficient volume of reviews is needed, and this requires going beyond the strict extraction limit of only 5 instances imposed by the standard Google Places API [21]. Data is collected from Google Maps through a semi-automated DOM scraping workflow using a custom JavaScript-based scraping utility [59]. Researchers manually navigate to each target establishment, load and scroll through the review section, and initiate or terminate the scraping process as needed. Once the scraper is running, the script automates the repetitive detection and extraction of review elements from the loaded DOM, including star ratings and review text, while also performing basic text cleaning and duplicate detection. The extracted records are exported into CSV files, with the output split into separate files when necessary to facilitate data handling and subsequent preprocessing.

2. **Normalization and Cleaning:** Noisy, unstructured text undergoes preprocessing to correct informal spelling, normalize special characters, and convert emojis to textual representations. Thus, ensuring clean and consistent data for model ingestion [53].
3. **Generative Aspect Taxonomy Induction:** Rather than relying on assumed, static industry categories, generative Large Language Models (LLMs) are used to discover recurring themes and implicit operational dimensions specific to the regional Davao City consumer context (e.g., [Wi-Fi Reliability], [Parking Availability]). This establishes a data-driven master dictionary [17].
4. **Aspect Extraction:** Because reviews frequently contain mixed sentiments directed

at different attributes, traditional document-level classification is insufficient. In this stage, the model isolates specific areview-aspect pairs (e.g., Taste, Price, Service, Ambiance) from the unstructured text. This serves as a prerequisite mechanism to break down complex sentences into granular, analyzable targets before any sentiment is evaluated [55].

5. **LLM Synthetic Augmentation:** To catch class imbalances early in the pipeline, a data sufficiency threshold checks the class distribution across the isolated aspect dimensions right after extraction. Rather than relying on a fixed, arbitrary target, which often lacks empirical grounding in specific ABSA contexts, the threshold is determined empirically instead. Using a progressive sampling approach [50], a small subset of authentic data is trained incrementally. The model is first fine tuned on an initial batch of 50 instances, then expanded to 100, and then to 150. After each batch, the F1-score is evaluated and plotted to build a learning curve. This process reveals how much data the model actually needs to learn the patterns of a specific category. The point where adding more data stops producing meaningful gains, where the curve levels off, becomes the minimum viable threshold for that class [26][47].

To avoid model collapse [22], the threshold acts as a dynamic ceiling instead of a hard cutoff. When a minority category drops below its baseline, the pipeline triggers Gemini 3 to synthesize more samples. Following the stabilization framework from He et al. [23], the system caps this generation at 30% of the total training volume. The process stops the moment the class hits the empirical threshold, matches the median frequency of natural classes [42], or reaches the 30% limit. Whichever condition happens first stops the script. This setup prevents runaway augmentation and mathematically blocks generative degradation.

6. **Active Learning and Manual Labeling:** To overcome the bottleneck of massive manual annotation and verify the quality of the dataset, the pipeline deploys the EASE framework [6]. Instead of randomly labeling data, the system uses informative and diverse sampling to selectively flag highly uncertain and non-redundant textual edge cases. These filtered samples are then routed to human annotators who perform manual labeling, acting as the "oracle" to provide definitive ground-truth sentiment labels for the most ambiguous code-switched text while drastically reducing overall human effort.
7. **Multilingual Transformer Fine-Tuning:** A pre-trained Cross-lingual Language Model (XLM-RoBERTa) serves as the core linguistic engine. In this phase, the model is actively fine-tuned using the study's constructed dataset to accurately interpret the contextual relationships and complex syntax of the localized, code-switched text. Building on the framework established by Maceda et al. [36], this study introduces several architectural modifications. Where the baseline study used a static, manually annotated dataset for general ternary sentiment classification, this pipeline works with a dynamic, hybrid dataset that combines human-annotated edge cases and strictly regulated LLM-generated synthetic data.
8. **Classification (Aspect-Level Sentiment Prediction):** Once the target aspects

have been extracted and the model fine-tuned, the pipeline evaluates the emotional polarity attached to each individual feature [55]. The model classifies the sentiment specific to that isolated review-aspect pair, enabling a highly nuanced and granular understanding of consumer feedback [53][7].

3.3 Output Phase

The output phase represents the final deliverables after the process. This will present Aspect-level polarities, Consumer insights, and Local Market Intelligence, transforming raw, unstructured feedback into metrics. Relying on an ambiguous average star rating often fails to provide specific operational insights. Extraction of granular aspect-level polarities allows consumers to compare establishments [28]. These extracted insights provide businesses with market intelligence enabling them to pinpoint operational strengths and address targeted areas for improvement [39] [53].

The conceptual framework demonstrated how challenges of processing large volumes of unstructured, code-switched local review can be effectively managed. Routing raw text through a robust and detailed pipeline of active learning, augmentation, and transformer-based aspect classification. It bridges the gap between online reviews and structured, actionable intelligence. Further discussions of technical configurations, procedures, and executions will be detailed in the succeeding methodology.

Chapter 4

Methodology

4.1 Research Design

The research follows a multi-stage computational pipeline designed to perform Aspect-Based Sentiment Analysis (ABSA) on code-switched consumer feedback. The architecture integrates transformer-based deep learning, generative artificial intelligence for taxonomy discovery, and human-in-the-loop active learning to provide granular market intelligence.

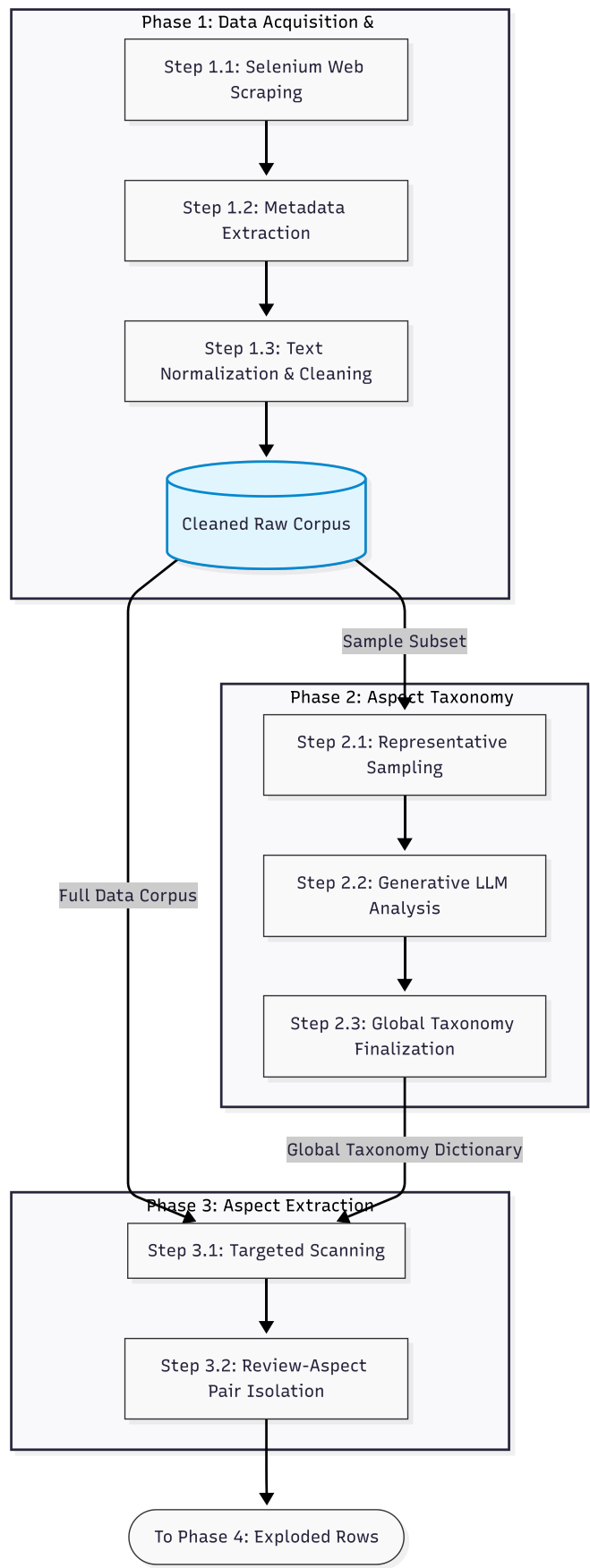


Figure 10: *Data Acquisition and Aspect Extraction*

4.1.1 Phase 1: Data Acquisition and Textual Preprocessing

The initial phase, as shown in Figure 10, is the automated harvesting and textual normalization of the primary research corpus.

- **Step 1.1: Semi-Automated DOM Scraping:** Bypassed Google Places API restrictions using a custom JavaScript scraper executed directly in the browser console. Researchers manually navigate to a venue’s review tab and run the script while scrolling to trigger lazy loading. As new content renders, the script continuously extracts the coffee shop’s name, geographical coordinates, star ratings, and raw review text. It handles data persistence in the background, caching the extracted nodes in memory before the virtual DOM unmounts them. The researcher monitors the feed and halts execution when the reviews run out or the page layout breaks. The script then compiles the persisted data and exports it directly into a CSV file for that specific establishment.
- **Step 1.2: Data Extraction:** For each retrieved entry, the system extracts the raw review text including numerical star ratings, geographic coordinates, and establishment identifiers.
- **Step 1.3: Textual Normalization:** Utilizing the *Pandas* library and Regular Expressions (Regex), the corpus undergoes rigorous cleaning. This process involves the removal of HTML artifacts, normalization of idiosyncratic characters, and the conversion of emoticons into descriptive text to preserve semantic sentiment cues during model training.

4.1.2 Phase 2: Aspect Taxonomy

Rather than adopting a static industry standard taxonomy or relying on traditional unsupervised topic modeling like Latent Dirichlet Allocation (LDA), which already tends to struggle with the noisy, informal syntax of code-switched regional text, this research uses a direct generative approach to identify localized operational dimensions instead. Recent work by Boughanmi et al. [9] shows that large language models can directly produce semantic taxonomies that match or even outperform human coding, without needing an intermediate LDA step.

- **Step 2.1: Representative Stratified Sampling:** A subset of normalized reviews is randomly sampled to serve as a representative seed for exploratory analysis.
- **Step 2.2: Generative Semantic Analysis:** The sample is processed via a Generative Large Language Model (Gemini 3). The prompt design explicitly instructs the model to act as a qualitative researcher, tasked with identifying recurring linguistic patterns, frequent noun phrases, and implicit themes specific to the regional Davao City consumer context (e.g., isolating Bislish terms for taste or ambiance).
- **Step 2.3: Global Taxonomy Formalization and Validation:** A master dictionary of aspects is generated based on the LLM output. Before deployment, this dictionary undergoes a human-in-the-loop validation check by the researchers to

ensure context-specific dimensions, such as [Parking Availability] and [Wi-Fi Reliability], are accurately captured alongside traditional metrics.

4.1.3 Phase 3: Granular Aspect Extraction and Data Decomposition

To achieve high-resolution sentiment mapping, unstructured feedback is decomposed into granular units. Refer to Figure 10.

- **Step 3.1: Targeted Scanning:** An extraction algorithm utilizes the formal taxonomy to scan the comprehensive review dataset for explicit and implicit aspect mentions.
- **Step 3.2: Review-Aspect Pair Isolation:** This process, referred to as the "Data Explosion," partitions multi-aspect reviews into structured rows. Each row maps a specific textual segment to a single target aspect (e.g., $[Text] + [TargetAspect]$). This prevents "sentiment cancellation," in which a single review shows conflicting polarities across different operational dimensions.

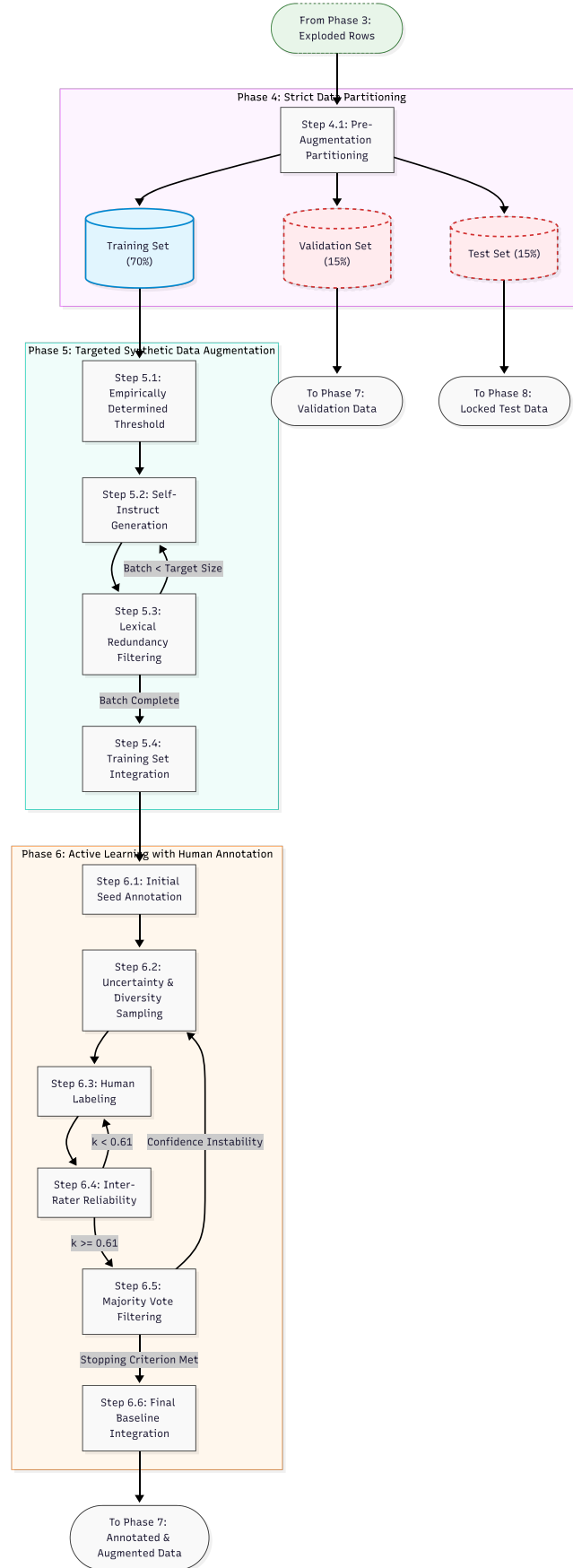


Figure 11: Partitioning, Augmentation and Annotation

4.1.4 Phase 4: Strict Data Partitioning

To maintain experimental integrity and preclude data leakage, strict partitioning is enforced, as shown in Figure 11.

- **Step 4.1: Pre-Augmentation Partitioning:** The authentic dataset is partitioned into Training (70%), Validation (15%), and Test (15%) sets. This was done to prevent data leakage and maintain experimental integrity. Additionally, a composite variable combining the target aspect and the sentiment polarity (e.g., "Customer Service_1-Star") is generated before splitting to ensure class imbalances are proportionally distributed across all sets, preventing a scenario where the Test set entirely lacks minority negative sentiments.
- **Step 4.2: Evaluation Firewall:** The Validation and Test sets are isolated from all synthetic data and training iterations. This ensures that final performance metrics represent the model's genuine generalizability to authentic human language.

4.1.5 Phase 5: Targeted Synthetic Data Augmentation

To calculate sufficient data thresholds. The dataset in phase 5, which originally has 1-to-5 star ratings are converted into proxy labels (e.g 1-2 stars mapped "Negative"). These proxy labels are used exclusively to train a TF-IDF and Logistic Regression baseline to plot learning curves. To avoid class imbalances found in niche operational categories, synthetic data augmentation is strictly utilized as a fallback mechanism, shown in Figure 11.

- **Step 5.1: Empirically Determined Threshold (Learning Curve):** Rather than setting a fixed, arbitrary threshold for data sufficiency, the minimum number of instances needed for each aspect category pair will be determined empirically. Using the weakly labeled corpus, the baseline model is trained on subsampled batches up to 10,000 total instances [50], and the resulting F1 scores will be plotted to produce a learning curve. The point where adding more data stops meaningfully improving performance, where the curve levels off, is what will define the minimum viable threshold for that specific class [26][47].
- **Step 5.2: Self-Instruct Generation:** If a specific category and star rating remain underrepresented and cannot naturally meet the empirically determined threshold, a Large Language Model (Gemini 3) will be utilized. It will use the Self-Instruct framework to synthesize additional training samples. Authentic reviews serve as "seed context" to ensure that the synthetic output accurately replicates the regional Bislish and Taglish syntactic nuances.
- **Step 5.3: Lexical Redundancy Filtering (TF-IDF):** Synthetic data generation often produces repetitive phrasing that dilutes the dataset. To enforce true lexical variance, candidate synthetic rows are transformed into sparse feature vectors using the TF-IDF vectorizer from Step 5.1. Pairwise cosine similarity is then computed both across the new synthetic batch to isolate internal duplicates and against the authentic baseline data. If a synthetic instance reaches a similarity score of ≥ 0.85 , it is discarded. Because this pruning shrinks the batch below target dimensions, the

generation pipeline operates as a continuous loop. Exact class deficits are recalculated, and subsequent generation prompts are updated with negative constraints, explicitly banning the n-grams of the rejected text. This cycle repeats until the precise volume of structurally distinct Bislish and Taglish reviews required for the final dataset is achieved.

- **Step 5.4: Training Set Integration:** Synthetic data is merged exclusively into the Training set, resulting in a balanced corpus that prevents model bias toward high-frequency categories while protecting the integrity of the validation sets.

4.1.6 Phase 6: Active Learning with Human Annotation

Annotation efficiency is maximized through an iterative labeling process (see Figure 11).

- **Step 6.1: Initial Seed Annotation:** Since distant supervision proxy labels utilized in Phase 5 contain inherent noise (e.g., a five-star review expressing a specific operational complaint), manual intervention is strictly required to establish definitive ground truth. Four (4) Domain-Competent Peer Annotators, comprising undergraduate students from Ateneo de Davao University who are native speakers of regional Bislish and Taglish dialects, will review a seed batch of the authentic training data. Rather than labeling from a blank slate, their task is to validate or correct the weak proxy labels. Their localized linguistic fluency qualifies them to accurately interpret highly contextual code-switching, thereby establishing the clean baseline required to initialize the XLM-RoBERTa active learning querying engine.
- **Step 6.2: Uncertainty and Diversity Sampling:** A preliminary baseline instantiation of the XLM-RoBERTa model is trained on the seed data to act as the active learning querying engine. This in-loop model flags "low-confidence" predictions (where the probability margin between the top predicted classes is minimal) and non-redundant samples [42]. These instances are prioritized for human intervention, ensuring that manual labeling efforts are focused strictly on the most ambiguous segments rather than redundant data [6].
- **Step 6.3: Human Labeling:** The flagged edge cases are routed back to the Domain-Competent Peer Annotators for a targeted round of manual annotation.
- **Step 6.4: Inter-Rater Reliability (Fleiss' Kappa):** The overall consistency of the annotations is mathematically evaluated using the Fleiss' Kappa statistic. To ensure high-quality ground truth, the study aims for a rigorous standard. If the batch achieves a score of $k \geq 0.61$ (Substantial Agreement) [32], it is validated and proceeds to the next step. If a batch falls below this threshold, it triggers a mandatory recalibration phase where annotators must review discordant instances, refine guidelines, and re-annotate until the substantial reliability threshold is met.
- **Step 6.5: Majority Vote Filtering:** For the mathematically validated batch, individual ground-truth labels are finalized using a strict majority-voting mechanism. Instances of absolute disagreement are considered highly ambiguous and permanently discarded to prevent noise from being introduced during the fine-tuning phase.

- **Step 6.6: Final Baseline Integration for Phase 7:** Upon meeting the active learning stopping criterion, the final human-annotated authentic dataset is merged with the targeted proxy-labeled synthetic instances generated in Phase 5. This final integration strictly caps the synthetic influence at 30% of the overall training volume. This step acts as a final structural safeguard, resolving any remaining minority class deficits without overriding the authentic linguistic feature space [23], thereby establishing a balanced and defensible baseline for Phase 7 model training.

Since active learning runs as an iterative cycle, a stopping criterion is enforced to keep the process from querying indefinitely. The annotation loop repeats until the model reaches Confidence Stability, which is the point where the remaining unlabeled pool no longer produces instances that fall below the uncertainty threshold. As a logistical fail-safe, the loop will also stop if it hits a predefined maximum annotation budget, essentially a hard cap on the total number of instances based on the time and availability of the volunteer peer annotators. Once either condition is met, the active learning phase ends and the dataset moves to Phase 7.

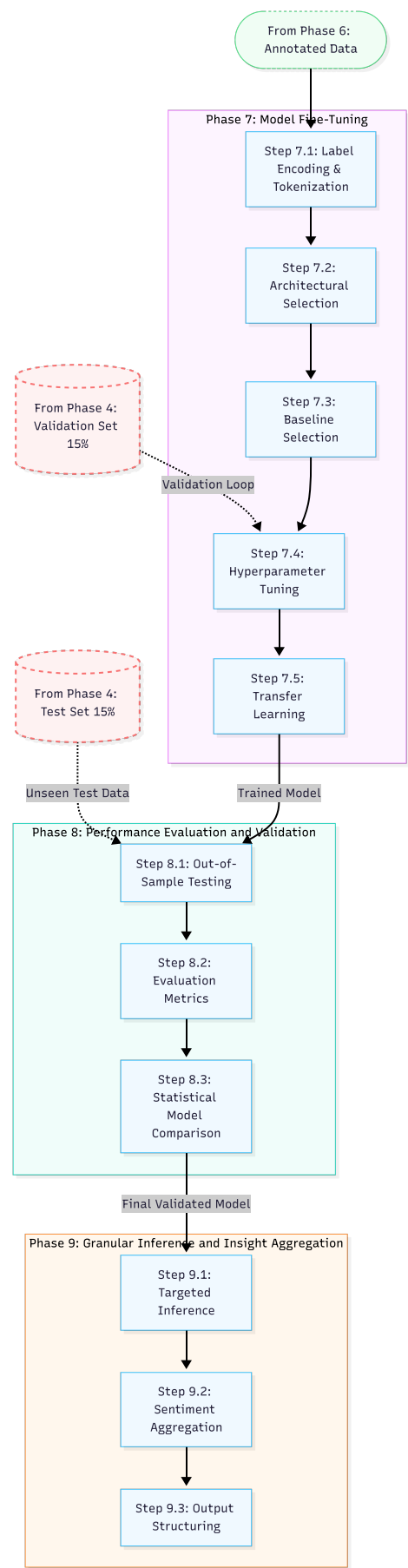


Figure 12: *Modeling and Evaluation*

4.1.7 Experimental Setup and Reproducibility

Hardware and Software Environment: All computational experiments, data pre-processing, and model fine-tuning are executed on a custom PC machine equipped with a Ryzen 7 5800x processor, 16GB of RAM, and an NVIDIA RTX 3050 (8GB VRAM) GPU. The software environment operates on Windows 11 using Python 3.10. The core machine learning pipeline relies on PyTorch (v2.14.0) as the primary tensor backend and the Hugging Face Transformers library (v5.17.0) for model architecture and tokenizer implementations. Data manipulation and numerical operations are handled via Pandas (v3.0.5) and NumPy (v2.5.1), while Scikit-Learn (v1.9.0) is utilized for baseline traditional machine learning models and metric evaluations.

Version Control: To ensure full transparency and reproducibility, the source code, dataset samples, and preprocessing scripts are version-controlled and publicly accessible via a GitHub repository (<https://github.com/z-sulit/Thesis-Project-Aspect-Based-Sentiment-Analysis>). The repository includes a comprehensive `README.md` detailing dependency installation via `requirements.txt` and execution instructions for running a minimal working example of the fine-tuning pipeline.

4.1.8 Phase 7: Model Fine-Tuning

The core sentiment engine is optimized for the code-switched linguistic landscape of the Philippines (see Figure 12).

- **Step 7.1: Label Encoding and Tokenization:** The pipeline sets the data for PyTorch by mapping the sentiment labels directly to integers: Positive as 0, Neutral as 1, and Negative as 2. This step is strictly required to calculate Cross-Entropy Loss. Alongside the label encoding, the XLM-RoBERTa tokenizer parses the raw code-switched reviews, converting the text strings into the padded numerical tensor sequences the model expects.
- **Step 7.2: Architectural Selection:** The research utilizes the *XLM-RoBERTa-base* architecture via the *Hugging Face Transformers* library. XLM-RoBERTa is selected over mBERT due to its superior performance on informal web-scraped text and its extensive pre-training on a 2.5TB corpus that includes Cebuano and Tagalog [51].
- **Step 7.3: Baseline Selection:** To rigorously quantify the performance gains of the proposed architecture, XLM-RoBERTa is benchmarked against two standard baselines established in the related literature. First, a Support Vector Machine (SVM) with TF-IDF is used as the traditional machine learning baseline based on the framework of Suciati and Budi [53]. Second, Multilingual BERT (mBERT) is used as the primary transformer baseline following the methodology of Maceda et al. [36], determining whether XLM-RoBERTa’s larger pre-training corpus provides a statistically significant advantage.
- **Step 7.4: Hyperparameter Tuning Strategy:** Optimizing the XLM-RoBERTa model requires balancing convergence speed against the risk of catastrophic forgetting [14] [37]. The tuning strategy utilizes a Grid Search method over a constrained

parameter space. The number of training epochs is fixed at 5 with an Early Stopping patience of 2 to prevent overfitting on the synthetic data injections, and weight decay is fixed at 0.01 for regularization [35]. The learning rate is searched across $\{2e-5, 3e-5, 5e-5\}$ and the batch size is tested at $\{16, 32\}$ following established transformer fine-tuning ranges [37], optimized to fit within the VRAM constraints of the hardware environment.

- **Step 7.5: Transfer Learning:** The model undergoes fine-tuning on a training set, learning complex code-switched syntax with specific aspect-level sentiment polarities.

4.1.9 Phase 8: Performance Evaluation and Validation

The pipeline’s effectiveness is validated against the actual data (see Figure 12).

- **Step 8.1: Out-of-Sample Testing:** The fine-tuned XLM-RoBERTa model is evaluated using the locked Test set, which contains 100% authentic, unseen human feedback.
- **Step 8.2: Evaluation Metrics:** Model performance is evaluated using Precision, Recall, and the F1-Score [64]. Given the inherent class imbalance of consumer review data, the Macro-F1 score is the primary evaluation metric. Macro-averaging computes the metric independently for each class and then takes the unweighted mean, ensuring the model is penalized if it defaults to predicting the dominant positive sentiment class while failing on the minority negative classes [54].
- **Step 8.3: Statistical Model Comparison:** To determine if the performance difference between XLM-RoBERTa and the baseline mBERT model is statistically significant, the study employs McNemar’s Test. This non-parametric test is specifically designed for comparing two classifiers evaluated on the same nominal test set. It isolates the discordant predictions (cases where one model predicts correctly and the other fails) to calculate the test statistic [16]. The equation is defined as:

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

Where b represents the number of test instances where XLM-RoBERTa was correct and mBERT was incorrect, and c represents the number of instances where mBERT was correct and XLM-RoBERTa was incorrect. A significance level of $\alpha = 0.05$ will be used to reject the null hypothesis.

4.1.10 Phase 9: Granular Inference and Insight Aggregation

The final phase translates the computational outputs into structured, actionable insights designed to support decision-making for both consumers and business owners (see Figure 12).

- **Step 9.1: Targeted Inference:** The validated model processes authentic feedback for individual target establishments to generate aspect-level sentiment predictions.

- **Step 9.2: Sentiment Aggregation:** The predicted polarities are compiled to provide a comprehensive, quantitative overview of consumer satisfaction. The system aggregates the counts of positive, negative, and neutral sentiments for each discovered operational dimension and calculates their prevailing sentiment percentages (e.g., determining that 85% of reviews mentioning “Wi-Fi Reliability” are positive).
- **Step 9.3: Output Structuring (Granular System Interface):** Rather than returning raw computational output, the final vision for system integration is to present these aggregated metrics in a structured, consumer facing format. The system will display a directory-style listing of local coffee and matcha establishments, replacing vague average star ratings (4.2 stars, 4.0 stars) with granular, aspect-specific scores. A specific shop might show a 75% positive sentiment for Taste but a 60% negative sentiment for Parking Availability. Following the principle of translating model outputs into something practically useful and easy to read, this structured output works as a proof of concept.

4.2 Data Collection

The primary dataset for this study consists of publicly available consumer reviews of coffee and matcha establishments in Davao City, collected from Google Maps. Although the Google Places API offers a structured way to retrieve this kind of data, it limits results to roughly five reviews per establishment. Chen et al. (2024) [12] and Trezza (2023) [57] both noted that this restriction makes APIs impractical for training machine learning models, and this study encountered the same limitation. A custom web scraper was developed to collect sufficient review data to fine-tune the XLM-RoBERTa model (Figure 13).

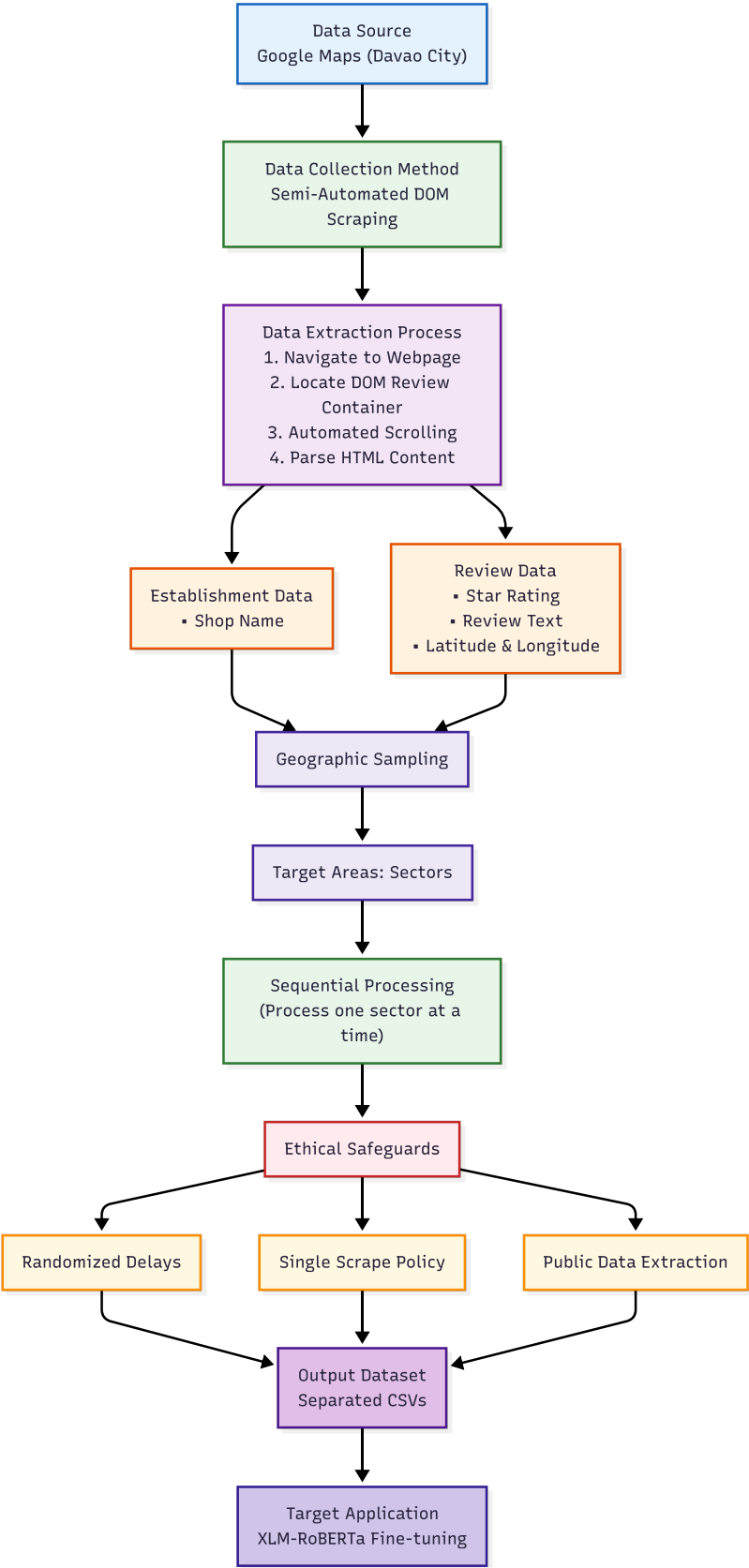


Figure 13: *Data Collection Process*

4.2.1 Extraction Architecture and Logic

Data collection was carried out using custom DOM JavaScript and Python web scraping scripts. Google Maps loads its content dynamically through JavaScript and utilizes DOM virtualization. To bypass this, a semi-automated DOM scraping approach was implemented. The review pane was manually scrolled to trigger the network requests and force the lazy-loaded content to render fully within the browser. Once the DOM was populated, custom JavaScript routines extracted the target shop’s name, overall rating, and granular data for each entry, including the reviewer’s name, star rating, full review text, and publication date. The extracted payload was then passed to a Python script for structuring. To maintain dataset organization, the resulting structured data was partitioned and exported into multiple CSV files, strictly categorized by business type (e.g., separating Coffee Shops and Matcha Shops into distinct files).

4.2.2 Systematic Geographic Sampling

To cover Davao City without making the data collection process unmanageable, the study divided the city into key commercial sectors such as Ecoland, Matina Crossing, Poblacion District, and others. The extraction process targeted all identified coffee and matcha shops within one sector at a time before moving on to the next, which kept the process organized and reduced the risk of geographic gaps in coverage.

4.2.3 Ethical Considerations and Technical Safeguards

Automated scraping can place unintended loads on servers and trigger IP blocks if executed recklessly. Therefore, the data collection followed polite scraping principles. The semi-automated approach naturally introduced human-like time delays during the manual scrolling phase, avoiding the rapid, predictable request patterns that trigger anti-bot protections. Each establishment was scraped only once rather than on a recurring schedule to avoid straining the host servers. All collected data was publicly visible on Google Maps and was utilized strictly for this academic study with no commercial intent.

4.3 Data Description

4.3.1 Raw Data

Volume and Attrition

The master dataset contains 19,586 total reviews across 400 unique establishments, split between 17,454 coffee shop reviews across 349 locations and 2,132 matcha shop reviews across 51 locations. A significant attrition rate is present, as 7,015 instances (6,350 coffee; 665 matcha) are rating-only entries entirely lacking text. These empty entries are systematically dropped during preprocessing, yielding a final usable textual corpus of 12,571 reviews. Furthermore, the raw corpus extracted via the custom Semi-Automated DOM Scraping is inherently susceptible to sampling gaps and selection bias.

Baseline Distribution Skew

Response bias was observed in the dataset. The frequency of 5-star ratings vastly outnumbers all other categories combined, while 1-star, 2-star, and 3-star ratings are severely underrepresented. This massive deficit in negative baseline data establishes the empirical need for the targeted LLM synthetic augmentation phase later in the pipeline to prevent model bias.

Vocabulary, Bigrams & Linguistic Metrics

Extreme outliers highly skew review lengths. Coffee reviews average 20.4 ± 37.8 words, reaching a maximum of 619 words, while matcha reviews average 13.7 ± 19.4 words, peaking at 325 words. Despite these maximums, both categories share a median length of exactly 6 words.

Code-switching is a consistent feature, maintaining a 5.04% overall rate across the combined corpus (5.05% in coffee; 4.97% in matcha). The top extracted bigrams reflect broad, generic sentiment markers rather than specific operational feedback, dominated by phrases such as "good food" (462 occurrences), "nice place" (393 occurrences), and "best matcha" (58 occurrences).

The corpus contains a high frequency of sentiment-critical emojis, led by the red heart (378 occurrences), star (227 occurrences), and thumbs-up (219 occurrences). These elements undergo systematic demojization during the normalization phase, converting symbols into standardized text tokens to preserve semantic intent before transformer fine-tuning.

Table 1: Dataset Overview Summary

Metric Description	Coffee Shops	Matcha Shops	Combined Corpus
Total Master Reviews	17,454	2,132	19,586
Reviews with Text	11,104	1,467	12,571
Rating-Only Reviews (Empty Text)	6,350	665	7,015
Unique Establishments	349	51	400

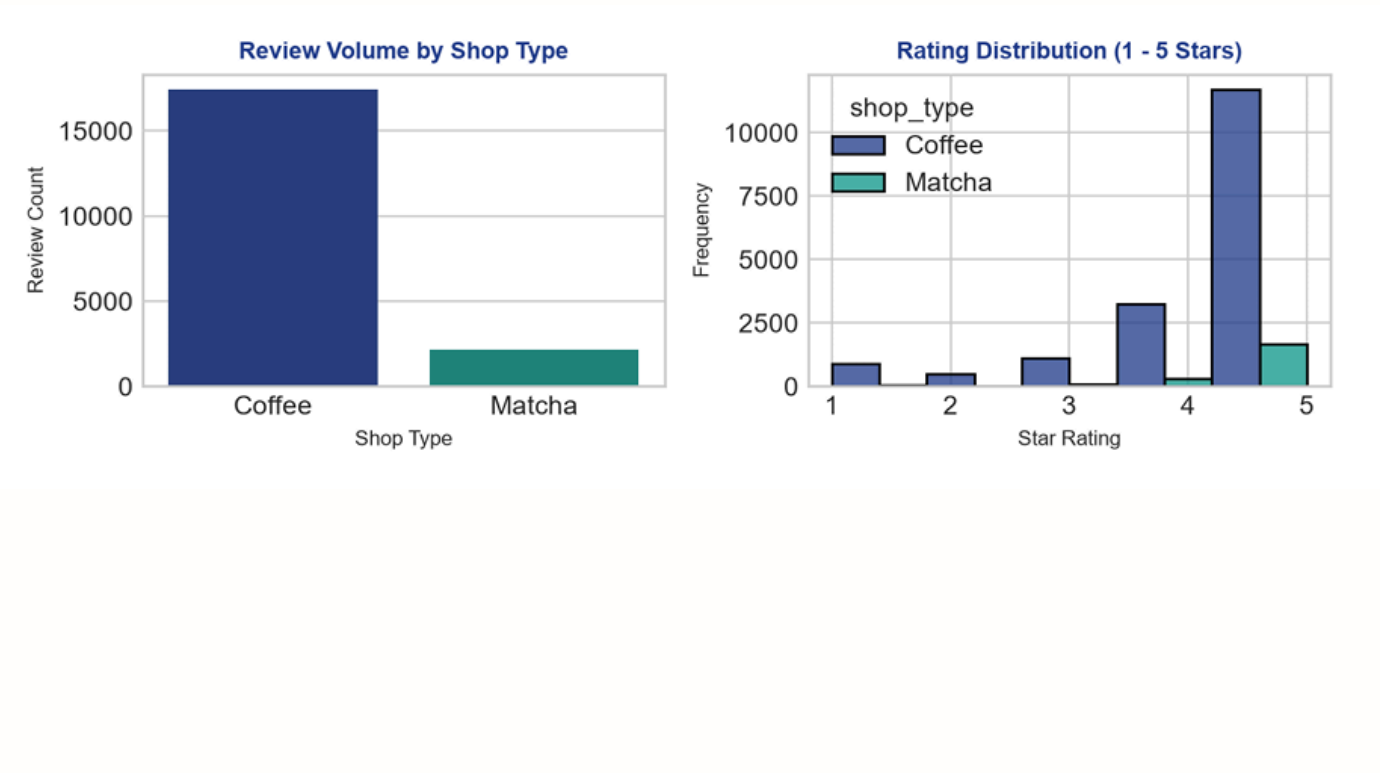


Figure 14: Dataset Volume and Rating Distribution Skew

Table 2: Review Length and Word Count Profiling

Shop Type	Count	Mean $\pm$ Std	Median (50%)	75th Pct	Max Outlier
Coffee	17,454	20.4 $\pm$ 37.8	6 words	24 words	619 words
Matcha	2,132	13.7 $\pm$ 19.4	6 words	21 words	325 words

Table 3: Vocabulary, Bigrams & Linguistic Metrics

Category	Top Bigrams (Count)	Type-Token Ratio (TTR)	Code-Switching %
Coffee	good food (462), coffee shop (430), nice place (393), food good (345), good place (273)	0.0823	5.05%
Matcha	matcha latte (73), best matcha (58), love matcha (54), good matcha (50), matcha good (39)	0.1718	4.97%
Overall	Combined corpus bigram profiling	0.0792	5.04%

Table 4: Top Frequency Emoji Analysis

Rank	Demojized Name	Occurrences
#1	:red_heart:	378
#2	:star:	227
#3	:thumbs_up:	219
#4	:smiling_face_with_smiling_eyes:	169
#5	:smiling_face_with_heart-eyes:	165
#6	:hot_beverage:	156
#7	:hundred_points:	125
#8	:face_savoring_food:	124
#9	:smiling_face_with_hearts:	120
#10	:sparkles:	108

4.3.2 Review Aspect Pairs Data

Class Imbalance

As detailed in Table 5, the dataset exhibits severe class imbalance, with Product Quality dominating at 47.88%, while Price and Value (5.18%) and Store Operations (5.24%) are severely underrepresented. Training a model on this raw distribution will cause it to overpredict the majority class and fail on the minority categories. This validates the absolute necessity of executing Phase 5 (Targeted Synthetic Data Augmentation) to generate targeted text for these specific deficits before moving into the active learning loop.

Validation of the De-composition Strategy

Because over 54% of the scraped reviews mention two or more distinct operational aspects (Table 6), standard document-level sentiment analysis would have failed on this corpus. This high aspect density proves that the Phase 3 decision to explode the reviews into granular, sentence-level chunks was necessary for isolating targeted sentiment.

Volatility in Customer Service

While "Ambiance and Atmosphere" holds the highest average rating (4.46), Table 7 demonstrates that "Customer Service" is highly polarizing. It has the lowest mean rating (4.00) alongside the highest standard deviation (1.444), indicating that customers primarily mention service when it fails.

Table 5: Aspect Class Imbalance Summary

Target Aspect	Count	Percentage (%)
Product Quality	11,040	47.88%
Customer Service	4,611	20.00%
Ambiance and Atmosphere	3,085	13.38%
Facilities and Amenities	1,919	8.32%
Store Operations and Accessibility	1,208	5.24%
Price and Value	1,195	5.18%

Table 6: Aspects by Shop Type (Frequencies and Percentages)

Target Aspect	Coffee (Count)	Matcha (Count)	Coffee (%)	Matcha (%)
Product Quality	9,704	1,336	46.64%	59.30%
Customer Service	4,206	405	20.22%	17.98%
Ambiance and Atmosphere	2,799	286	13.45%	12.69%
Facilities and Amenities	1,820	99	8.75%	4.39%
Store Operations and Accessibility	1,155	53	5.55%	2.35%
Price and Value	1,121	74	5.39%	3.28%

Table 7: Overall Rating vs. Target Aspect (Sentiment Proxy)

Target Aspect	Count	Mean	Median	Std. Dev.
Ambiance and Atmosphere	3,085	4.463	5.0	0.911
Product Quality	11,040	4.286	5.0	1.171
Facilities and Amenities	1,919	4.138	5.0	1.196
Price and Value	1,195	4.035	5.0	1.317
Store Operations and Accessibility	1,208	4.028	4.0	1.238
Customer Service	4,611	4.005	5.0	1.444

4.3.3 Synthetic Data Augmentation and Ratio Validation

Augmented Training Baseline Distribution (Proxy-Labelled Sentiment)

(Table 8) demonstrates a successful hit the empirical threshold set in Phase 5.1. The system stopped generating the exact moment the underrepresented minority classes reached their minimum required volume. This hard cap leaves the Negative sentiment class with exactly 250 instances across four categories: Ambiance and Atmosphere, Facilities and Amenities, Price and Value, and Store Operations and Accessibility.

(Table 9) confirms the targeted augmentation prevented distribution shift and model collapse. By cutting off generation at the empirical threshold, the pipeline kept the synthetic text from washing out the original dataset. At most, synthetic data accounts for just 13.94% of the Store Operations and Accessibility category, leaving over 86% of the semantic signal rooted in authentic Bislish and Taglish syntax.

Synthetic Ratio Validation

The pipeline skipped generating data for Customer Service and Product Quality entirely, leaving both with a 0.0% synthetic ratio. Based on (Table 8), the natural minority classes for these categories (616 and 836 instances) already cleared the 250-instance target. The system acted purely as a safety net, stepping in only when mathematically necessary to keep the original corpus intact.

Table 8: Augmented Training Baseline Distribution (Proxy-Labelled Sentiment)

Target Aspect	# Negative	# Neutral	# Positive	# All
Ambiance and Atmosphere	250	144	1903	2297
Customer Service	616	244	2368	3228
Facilities and Amenities	250	130	1057	1437
Price and Value	250	102	610	962
Product Quality	836	570	6321	7727
Store Operations and Accessibility	250	80	653	983
All	2452	1270	12912	16634

Table 9: Synthetic Ratio Validation

Target Aspect	Synthetic Data %
Ambiance and Atmosphere	6.01%
Customer Service	0.0%
Facilities and Amenities	6.47%
Price and Value	13.1%
Product Quality	0.0%
Store Operations and Accessibility	13.94%

4.3.4 Active Learning and Annotations

Table 10 shows the balanced structure produced by the Phase 6 baseline integration. By injecting 30% synthetic data, the pipeline boosted the critical minority classes such as Negative Ambiance, Facilities, Price, and Store Operations to a working range of 55 to 63 instances each. This gives the XLM-RoBERTa model enough statistical weight to map decision boundaries for rare complaints without overfitting.

The broader dataset distribution confirms the project avoided generative collapse. Positive sentiment still dominates with 409 total instances, keeping the model grounded in real Bislish and Taglish. Since they did not need the boost, Negative Customer Service and Product Quality were left untouched to minimize synthetic noise, holding their original counts of 14 and 21 instances. The Neutral class was also kept intentionally sparse, steering the model toward actionable business insights rather than flat observations.

Table 10: Cross-Tabulation: Target Aspect by Sentiment (Added Synthetic Data)

Target Aspect	Negative	Neutral	Positive	Total
Ambiance and Atmosphere	63	4	187	254
Customer Service	14	6	34	54
Facilities and Amenities	57	2	57	116
Price and Value	56	1	10	67
Product Quality	21	24	102	147
Store Operations and Accessibility	55	2	19	76
Total	266	39	409	714

4.4 Bias Mitigation and Data Integrity Controls

To ensure the authenticity of the XLM-RoBERTa model and prevent algorithmic bias during training, the dataset underwent strict control mechanisms before fine-tuning.

Strict Stratified Partitioning

The use of the 70/15/15 split implementation as an Evaluation Firewall. By locking the validation and test sets before any data processing, we can be assured that our performance metrics measure generalization to the original Taglish and Bislish text, free from the influence of training contamination.

Deduplication and Noise Filtering

Raw text corpora contain structural noise that skews transformer-based classification. To address this, the first preprocessing step dropped 7,015 rating-only entries. Since ABSA depends entirely on extracting operational dimensions from written text, empty strings offer no value. Leaving them forces the XLM-RoBERTa tokenizer to process zero-information vectors, thus wasting compute and distorting gradient updates during fine-tuning.

Empirical Class Thresholding

The use of progressive sampling to determine the threshold for the F1-score metric. It minimizes the potential problems caused by the introduction of synthetic noise that causes the issues encountered in overusing sampling of minority data classes to balance large majority classes. The values of 500 and 1,000 instances mentioned are actual empirical values obtained from results extracted from learning curves.

Validation Set Isolation

Synthetic data was injected exclusively into the Training set. The Validation and Test sets were strictly isolated to contain 100% authentic human text, completely preventing data leakage and ensuring untainted downstream evaluation metrics.

Synthetic Ratio Capping Imbalance Resolution

Synthetic data injection was capped at 30% of the training baseline to prevent generative collapse and keep authentic human text as the primary learning signal. Cross-tabulating target aspects by sentiment verified the augmentation, confirming the minority classes hit their necessary data thresholds without artificially inflating the dominant positive classes.

Cosine Similarity Collision Filtering

A TF-IDF Cosine Similarity filter with a drop threshold of ≥ 0.85 was applied during data augmentation. This efficiently detected and discarded synthetic outputs that regurgitated or heavily plagiarized existing authentic reviews, ensuring dataset diversity.

Margin and Diversity Sampling

During the active learning cycle, margin calculations were used to rank unannotated rows by predictive uncertainty. Aspect stratification (Diversity Sampling) was then applied to extract exactly 400 diverse, highly ambiguous rows for human annotations. These 400 rows were subsequently appended to the initial 100-row seed batch, optimizing the 500-row annotator budget to form the authentic ground-truth baseline. To resolve minority class deficits, Phase 6 integrated this 500-row authentic set with targeted synthetic data strictly capped at 30% of the overall volume (214 synthetic instances). This precise integration mathematically expanded the final training dataset to the 714 total instances detailed in Table 10.

Inter-Rater Reliability Adjudication

Four independent annotators evaluated the dataset to establish an unbiased ground truth for the authentic text. Before finalizing the labels, Fleiss' Kappa was calculated to verify inter-rater reliability. A majority-vote algorithm resolved standard disagreements. In the event of a direct tie, the lead researcher manually adjudicated the instance to keep arbitrary machine bias out of the training set.

Bibliography

- [1] ABLAZO, A. D. Designing a web application using the twitter streaming API as a tracking tool for weather updates. Ateneo de Davao University, 2019.
- [2] ADNAN, K., AND AKBAR, R. An analytical study of information extraction from unstructured and multidimensional big data. *Journal of Big Data* 6, 1 (2019), 1–38. Retrieved from: <https://link.springer.com/article/10.1186/s40537-019-0254-8>, Date retrieved: March 30, 2026.
- [3] ALAMA, R. I. Davao’s coffee and cacao experience: A platform of shared opportunities, oct 2023. Retrieved from: <https://pia.gov.ph/features/davaos-coffee-and-cacao-experience-a-platform-of-shared-opportunities/>, Date retrieved: April 02, 2026.
- [4] ALCARAZ, C. J., BELANISO, R. E., DAGAY, R., AND MELENDRES, A. G. Generation z tourists’ attitude, experience, and visit intention to local coffee shops in davao city. *La Ricerca* 10, 1 (2024). Retrieved from: <https://ieesjournals.com/index.php/la Ricerca/article/view/81>, Date retrieved: March 31, 2026.
- [5] ALMUAYQIL, S. N., HUMAYUN, M., JHANJHI, N., ALMUFAREH, M. F., AND KHAN, N. A. Enhancing sentiment analysis via random majority under-sampling with reduced time complexity for classifying tweet reviews. *Electronics* 11, 21 (2022). Retrieved from: <https://www.mdpi.com/2079-9292/11/21/3624>, Date retrieved: April 04, 2026.
- [6] ALTURAYEIF, N., AND AHMAD, I. Ease: An enhanced active learning framework for aspect-based sentiment analysis based on sample diversity and data augmentation. *Expert Systems with Applications* 261 (2025), 125525.
- [7] AMALIA, P. Aspect-based sentiment analysis on indonesian restaurant review using a combination of convolutional neural network and contextualized word embedding. *IJCCS (Indonesian Journal of Computing and Cybernetics Systems)* (2021). Retrieved from: <https://www.academia.edu/31842>, Date retrieved: March 30, 2026.
- [8] ANG NGO CHING, P. K. W., LIPARDO, D. R., CHICO, C. J. G., AND PEDRAJA, M. A. F. Analyzing the influence of language use on sentiment analyzers using reviews gathered from internet platforms. Retrieved from: https://animorepository.dlsu.edu.ph/conf_shsrescon/2025/paper_csr/5/, Date retrieved: April 03, 2026.

- [9] BOUGHANMI, K., JEDIDI, K., AND JEDIDI, N. From reviews to actionable insights: An llm-based approach for attribute and feature extraction. *arXiv preprint arXiv:2510.16551* (2025).
- [10] BRADY, A., AND ISLAM, T. Iterative topic taxonomy induction with llms: A case study of electoral advertising. *arXiv preprint arXiv:2510.15125* (2025).
- [11] CHAMID, A. A., KUSUMANINGRUM, R., ET AL. Labeling consistency test of multi-label data for aspect and sentiment classification using the cohen kappa method. *Ingenierie des Systemes d'Information* 29, 1 (2024), 161.
- [12] CHEN, Y., SHERREN, K., LEE, K. Y., MCCAY-PEET, L., XUE, S., AND SMIT, M. From theory to practice: insights and hurdles in collecting social media data for social science research. *Frontiers in Big Data* 7 (2024). Retrieved from: <https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2024.1379921/full>, Date retrieved: April 01, 2026.
- [13] CHIUSANO, F. Quick intro to Aspect-Based Sentiment Analysis, 2022. Retrieved from: <https://medium.com/nlplanet/quick-intro-to-aspect-based-sentiment-analysis-c8888a09eda7>, Date retrieved: April 01, 2026.
- [14] CONNEAU, A., KHANDLWAL, K., GOYAL, N., CHAUDHARY, V., WENZKE, G., GUZMÁN, F., GRAVE, E., OTT, M., ZETTEMAYER, L., AND STOYANOV, V. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (Online, July 2020), D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds., Association for Computational Linguistics, pp. 8440–8451.
- [15] COSME, C. J., AND DE LEON, M. M. Sentiment analysis of code-switched filipino-english product and service reviews using transformers-based large language models. In *World Conference on Information Systems for Business Management* (2023), pp. 123–135. Retrieved from: https://link.springer.com/chapter/10.1007/978-981-99-8349-0_11, Date retrieved: April 04, 2026.
- [16] DIETTERICH, T. G. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation* 10, 7 (1998), 1895–1923.
- [17] DIONELA, V. L. D., DY, F. K. S., HOMBREBUENO, R. J. M., NICOLAS, A. R. M., CHENG, C. K., AND GONDA, R. W. Aspect extraction from e-commerce product and service reviews. *arXiv preprint arXiv:2601.01827* (2026). Retrieved from: <https://arxiv.org/abs/2601.01827>, Date retrieved: April 05, 2026.
- [18] DONGO, I., CADINALE, Y., AGUILERA, A., MARTÍNEZ, F., QUINTERO, Y., AND BARRIOS, S. Web scraping versus twitter api: a comparison for a credibility analysis. In *Proceedings of the 22nd International conference on information integration and web-based applications & services* (2020). Retrieved from: <https://dl.acm.org/doi/abs/10.1145/3428757.3429104>, Date retrieved: March 31, 2026.
- [19] EL DEHAIBI, N., AND MACDONALD, E. Investigating inter-rater reliability of qualitative text annotations in machine learning datasets. In *Proceedings of the*

- Design Society: DESIGN Conference* (2020), vol. 1, Cambridge University Press, pp. 21–30.
- [20] GABATO, L. J., LATIP, A. S., PIELAGO, K. K., SANCHEZ, K. N., AND VILVESTRE, A. A conjoint analysis of the preferred digital marketing approaches for coffee shops among gen z college students in davao city. Retrieved from: <https://www.researchgate.net/publication/392324480>, Date retrieved: April 05, 2026.
 - [21] GOOGLE. Place details — Google Places API. <https://developers.google.com/maps/documentation/places/web-service/details>, 2026. Retrieved from: <https://developers.google.com/maps/documentation/places/web-service/details>, Date retrieved: April 02, 2026.
 - [22] GUO, Y., SHANG, G., VAZIRGIANNIS, M., AND CLAVEL, C. The curious decline of linguistic diversity: Training language models on synthetic text. In *Findings of the Association for Computational Linguistics: NAACL 2024* (2024), pp. 3589–3604. Retrieved from: <https://aclanthology.org/2024.findings-naacl.228/>, Date retrieved: April 01, 2026.
 - [23] HE, H., XU, S., AND CHENG, G. Recursive learning without collapse: a weighting-based stabilization framework. *Journal of the Royal Statistical Society Series B: Statistical Methodology* (2026), qkag099.
 - [24] HERHAUSEN, D., LUDWIG, S., ABEDIN, E., HAQUE, N. U., AND DE JONG, D. From words to insights: Text analysis in business research. *Journal of Business Research* 198 (2025). Retrieved from: <https://www.sciencedirect.com/science/article/pii/S0148296325003145>, Date retrieved: April 05, 2026.
 - [25] HERRERA, M., AICH, A., AND PARDE, N. Tweettaglish: A dataset for investigating tagalog-english code-switching. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference* (2022), pp. 2090–2097. Retrieved from: <https://aclanthology.org/2022.lrec-1.225/>, Date retrieved: April 04, 2026.
 - [26] HUANG, J., CUI, Y., LIU, J., AND LIU, M. Supervised and few-shot learning for aspect-based sentiment analysis of instruction prompt. *Electronics* 13, 10 (2024), 1924.
 - [27] HUTTO, C., AND GILBERT, E. Vader: A parsimonious rule-based model for sentiment analysis of social media text. In *Proceedings of the international AAAI conference on web and social media* (2014), vol. 8. Retrieved from: <https://ojs.aaai.org/index.php/icwsm/article/view/14550>, Date retrieved: April 03, 2026.
 - [28] INDEED EDITORIAL TEAM. Why are customer reviews so important? (and how to use them). *Indeed Career Guide* (2025). Retrieved from: <https://www.indeed.com/career-advice/career-development/why-are-customer-reviews-so-important>, Date retrieved: March 29, 2026.

- [29] INFINITY HUB. Digital marketing in Davao: Market analysis & opportunities, 2026. Retrieved from: <https://infinityhub.com/digital-marketing-davao-market-analysis-opportunities/>, Date retrieved: April 02, 2026.
- [30] JABR, W., AND RAHMAN, M. S. Online reviews and information overload: The role of selective, parsimonious, and concordant top reviews. *MIS quarterly* 46, 3 (2022). Retrieved from: <https://misq.umn.edu/misq/article-abstract/46/3/1517/1983/Online-Reviews-and-Information-Overload-The-Role>, Date retrieved: April 02, 2026.
- [31] JAYATHILAKE, H. D., YOON, A., AND HONG, M. Leveraging local-llm for sentiment analysis to enhance text data quality assessment in information science research. *The Electronic Library* (2026), 1–21. Retrieved from: <https://www.emerald.com/el/article-abstract/doi/10.1108/EL-08-2025-0354>, Date retrieved: April 04, 2026.
- [32] KRUSIC, L. Constructing a sentiment-annotated corpus of austrian historical newspapers: Challenges, tools, and annotator experience. In *Proceedings of the 4th International Conference on Natural Language Processing for Digital Humanities* (2024), pp. 51–62.
- [33] LEHTO, X., PARK, J. K., PARK, O., AND LEHTO, M. R. Text analysis of consumer reviews: The case of virtual travel firms. In *Symposium on Human Interface and the Management of Information* (2007), pp. 490–499. Retrieved from: https://link.springer.com/chapter/10.1007/978-3-540-73345-4_56, Date retrieved: April 02, 2026.
- [34] LI, G., WANG, A., WANG, S., ZHANG, Y., CAO, P., BAI, Y., AND LIU, C. H. Dall: Data labeling via data programming and active learning enhanced by large language models. *arXiv preprint arXiv:2602.14102* (2026).
- [35] LOSHCHILOV, I., AND HUTTER, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- [36] MACEDA, L. L., SATUITO, A. A., AND ABISADO, M. B. Sentiment analysis of code-mixed social media data on philippine uaqte using fine-tuned mbert model. *International Journal of Advanced Computer Science and Applications* 14, 7 (2023). Retrieved from: <https://pdfs.semanticscholar.org/9a13/b9aa1848908922bc35edd5f0c799dbdc8e94.pdf>, Date retrieved: April 02, 2026.
- [37] MALIK, M. S. I., YOUNAS, M. Z., JAMJOOM, M. M., AND IGNATOV, D. I. Categorization of tweets for damages: infrastructure and human damage assessment using fine-tuned bert model. *PeerJ Computer Science* 10 (2024), e1859.
- [38] MARGATINA, K. *Exploring Active Learning Algorithms for Data Efficient Language Models*. PhD thesis, University of Sheffield, 2024. Retrieved from: <https://theses.s.whiterose.ac.uk/id/eprint/35849/>, Date retrieved: April 03, 2026.
- [39] MATHAYOMCHAN, B., AND TAECHARUNGROJ, V. “how was your meal?” examining customer experience using google maps reviews. *International Journal of Hospitality*

- Management* 90 (2020), 102641. Retrieved from: <https://www.sciencedirect.com/science/article/abs/pii/S0278431920301936>, Date retrieved: April 01, 2026.
- [40] MONFARED, M. H. A., FLEK, L., AND KARIMI, A. Label-consistent data generation for aspect-based sentiment analysis using llm agents. In *The Proceedings for the 15th Workshop on Computational Approaches to Subjectivity, Sentiment Social Media Analysis (WASSA 2026)* (2026). Retrieved from: <https://aclanthology.org/2026.wassa-1.17/>, Date retrieved: April 02, 2026.
- [41] MUKHERJEE, A., AND LIU, B. Aspect extraction through semi-supervised modeling. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (ACL)* (Jeju, Republic of Korea, 2012), Association for Computational Linguistics, pp. 339–348.
- [42] MURALIDHAR, R., AND HULIPALLED, V. R. Sentiment analysis framework using deep active learning for smartphone aspect based rating prediction. *Foundations of Computing and Decision Sciences* 48 (2023). Retrieved from: <https://reference-global.com/download/article/10.2478/fcds-2023-0008.pdf>, Date retrieved: April 05, 2026.
- [43] OH, B.-D., KIM, G.-Y., KIM, C., AND KIM, Y.-S. How to use language models for synthetic text generation in cerebrovascular disease-specific medical reports. In *Proceedings of the 1st Workshop on Personalization of Generative AI Systems (PERSONALIZE 2024)* (2024), pp. 10–17.
- [44] ORTEGA, E. P. A bisaya language model for a neural-network based sentiment analyzer. *Animo Repository, De La Salle University, Manila* (2022). Retrieved from: https://animorepository.dlsu.edu.ph/etdm_softtech/5/, Date retrieved: March 29, 2026.
- [45] PAKPAHAN, M. M., DAR, M. H., AND HASIBUAN, M. N. S. Performance evaluation of machine learning algorithms in aspect-based sentiment analysis on e-commerce user reviews. *International Journal of Science, Technology & Management* 6, 4 (2025). Retrieved from: <https://ijstm.inarah.co.id/index.php/ijstm/article/view/1255>, Date retrieved: March 29, 2026.
- [46] PATIL, S. S., SURYAWANSHI, V. P., PATIL, S. M., GIRASE, S. P., AND BHAGAT, D. A. Review of sentiment analysis in social media using big data: techniques, tools, and frameworks. *International Journal of Basic and Applied Sciences* 14, 2 (2025), 34–48. Retrieved from: <https://www.researchgate.net/profile/Dhananjay-Bhagat/publication/392538982>, Date retrieved: April 02, 2026.
- [47] PONTIKI, M., GALANIS, D., PAPAGEORGIOU, H., ANDROUTSOPOULOS, I., AND MANANDHAR, S. Semeval-2014 task 4: Aspect based sentiment analysis. In *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)* (Dublin, Ireland, 2014), pp. 27–35.
- [48] PORIA, S., CAMBRIA, E., KU, L.-W., GUI, C., AND GELBUKH, A. A rule-based approach to aspect extraction from product reviews. In *Proceedings of the Second*

- Workshop on Natural Language Processing for Social Media (SocialNLP)* (Dublin, Ireland, 2014), Association for Computational Linguistics, pp. 28–37.
- [49] PRATMANTO, D., ROUSYATI, R., WATI, F. F., WIDODO, A. E., SULEMAN, S., AND WIJANTO, R. App review sentiment analysis shopee application in google play store using naive bayes algorithm. In *Journal of Physics: Conference Series* (2020), vol. 1641. Retrieved from: <https://iopscience.iop.org/article/10.1088/1742-6596/1641/1/012043/meta>, Date retrieved: April 05, 2026.
 - [50] PROVOST, F., JENSEN, D., AND OATES, T. Efficient progressive sampling. In *Proceedings of the fifth ACM SIGKDD international conference on Knowledge discovery and data mining* (1999), pp. 23–32.
 - [51] SALVE, H. N., AND TUBIL, P. N. T. Bilingual code-switching using xlm-roberta. *Authorea Preprints* (2025).
 - [52] SCHINDLER, B., AND RUZ, G. A. Geometric filtering of llm-generated samples for few-shot text classification. *arXiv preprint arXiv:2608.13866* (2026).
 - [53] SUCIATI, A., AND BUDI, I. Aspect-based sentiment analysis and emotion detection for code-mixed review. *International Journal of Advanced Computer Science and Applications* 11, 9 (2020). Retrieved from: <https://www.proquest.com/docview/2655132242>, Date retrieved: April 02, 2026.
 - [54] SUJON, K., HASSAN, R., CHOI, K., AND SAMAD, M. A. Accuracy, precision, recall, f1-score, or mcc? empirical evidence from advanced statistics, ml, and xai for evaluating business predictive models. *Journal of Big Data* 12 (12 2025).
 - [55] SURYABRATA, A. W., WARNARS, H. L. H. S., AND MUYEBA, M. K. Aspect-based sentiment analysis of multilingual hotel reviews in jakarta using multilingual transformer models. *Ingénierie des Systèmes d’Information* 31, 1 (2026). Retrieved from: <http://iieta.org/journals/isi>, Date retrieved: March 29, 2026.
 - [56] THARWAT, A., AND SCHENCK, W. A survey on active learning: State-of-the-art, practical challenges and research directions. *Mathematics* 11, 4 (2023), 820. Retrieved from: <https://www.mdpi.com/2227-7390/11/4/820>, Date retrieved: April 06, 2026.
 - [57] TREZZA, D. To scrape or not to scrape, this is dilemma. the post-api scenario and implications on digital research. *Frontiers in sociology* 8 (2023). Retrieved from: <https://www.frontiersin.org/journals/sociology/articles/10.3389/fsoc.2023.1145038/full>, Date retrieved: April 06, 2026.
 - [58] ULHAQ, A. L. D., AND SUPRAYOGI, S. Comparing machine learning models for sentiment analysis of tokopedia reviews. *Journal of Applied Informatics and Computing* 9, 6 (2025). Retrieved from: <https://jurnal.polibatam.ac.id/index.php/JAIC/article/view/11239>, Date retrieved: April 03, 2026.
 - [59] ULLOA, R., MANGOLD, F., SCHMIDT, F., GILSBACH, J., AND STIER, S. Beyond time delays: How web scraping distorts measures of online news consumption. *Communication Methods and Measures* 19, 3 (2025). Retrieved from: <https://>

- [//www.tandfonline.com/doi/full/10.1080/19312458.2025.2482538](https://www.tandfonline.com/doi/full/10.1080/19312458.2025.2482538), Date retrieved: April 03, 2026.
- [60] WALZ, R. What is sentiment analysis and big data sentiment analysis?, dec 2023. Retrieved from: <https://numerous.ai/blog/big-data-sentiment-analysis>, Date retrieved: April 06, 2026.
- [61] WANG, Y., KORDI, Y., MISHRA, S., LIU, A., SMITH, N. A., KHASHABI, D., AND HAJISHIRZI, H. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)* (2023), pp. 13484–13508. Retrieved from: <https://aclanthology.org/2023.acl-long.754/>, Date retrieved: April 05, 2026.
- [62] WINATA, G. I., CAHYAWIJAYA, S., LIU, Z., LIN, Z., MADOTTO, A., AND FUNG, P. Are multilingual models effective in code-switching? In *Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching* (2021). Retrieved from: <https://aclanthology.org/2021.calcs-1.20/>, Date retrieved: April 04, 2026.
- [63] WOLFF, J. U. A dictionary of cebuano visayan. cornell university. *Southeast Asia Program and Linguistic Society of the Philippines* (1972). Retrieved from: <https://www.gutenberg.org/files/40074/40074-h/40074-h.htm>, Date retrieved: April 06, 2026.
- [64] WU, Y., AND WAN, J. A survey of text classification based on pre-trained language model. *Neurocomputing* (February 2025).
- [65] ZHU, X., CHENG, D., LI, H., ZHANG, K., HUA, E., LV, X., DING, N., LIN, Z., ZHENG, Z., AND ZHOU, B. How to synthesize text data without model collapse? *arXiv preprint arXiv:2412.14689* (2024).